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A Rate-Splitting Strategy for STAR-RIS-Aided Massive MIMO Systems with Joint Optimization

Hanxiao Ge, *Graduate Student Member, IEEE*, Anastasios Papazafeiropoulos, *Senior Member, IEEE*, Navneet Garg, *Senior Member, IEEE*, and Tharmalingam Ratnarajah, *Senior Member, IEEE*

Abstract—This work proposes a rate-splitting (RS) strategy for simultaneously transmitting and reflecting reconfigurable intelligent surface (STAR-RIS) aided massive multiple-input multiple-output (mMIMO) systems to reduce the interference among multiple users and enhance the spectral efficiency (SE) while improving the coverage degraded by blockages. Specifically, we use the RS to design the precoder for the common part by solving the asymptotic problem. Also, unlike traditional RIS-aided systems, receivers can be positioned on either side of the RIS panel in the proposed system. We derive the sum-rate based on statistical channel state information (CSI) to reduce the signal overhead. Next, we optimize the rate through a projected gradient ascent method (PGAM) algorithm simultaneously with respect to the amplitudes and phase shifts of the STAR-RIS. Simulations show the advantages of the RS strategy compared to the broadcasting (BC) strategy in improving the sum-rate. We further evaluate the efficiency of the STAR-RIS system against the traditional RIS-aided system. In our analysis, we employ energy splitting (ES) and mode switching (MS) protocols to fine-tune the transmission and reflection coefficients of the outgoing and incoming signals.

Index Terms—Rate-splitting, STAR-RIS, massive MIMO, joint optimization, 6G communications.

I. INTRODUCTION

MASSIVE multiple-input multiple-output (mMIMO) has been a successful technology for cellular networks because it can enhance network throughput and energy efficiency [1], [2]. In parallel, reconfigurable intelligent surface (RIS), deployed with multiple reflective elements, has appeared as a promising technology for wireless systems. Each RIS element can change the phase shifts of the impinging signals, adjusted through a controller. Authors in [3] and [4] confirmed that the RIS can improve the spectral efficiency (SE) by reflecting the beams in a proper direction. Unfortunately, in most works concerning RIS, such as [3] and [5], the authors assumed that both the transmitter and receiver are situated on the identical side of the RIS. However, practical implementations may involve users on both sides, including the spaces in front and behind the surface. To address this issue, in works [6], [7], authors introduced the technology of simultaneously transmitting and reflecting RIS (STAR-RIS), which is of practical interest and improves the coverage and the SE compared to traditional RIS-aided systems. Authors in [7] applied the energy splitting (ES) and the mode switching (MS) protocols to adjust the transmission and reflection coefficients in STAR-RIS-aided systems.

In most RIS-aided systems, authors have considered the design with perfect channel state information (CSI) [8], [9]. This reduces the complexity but needs to meet realistic conditions. Also, it is hard to acquire any CSI because an RIS has passive elements and cannot transmit or receive any pilots

[10]. Hence, a more practical approach is to assume statistical and imperfect CSI. In this approach, channel estimation errors are inherently considered, making it a more robust and realistic framework for system design [4], [11]. Statistical CSI provides an estimated channel condition rather than the exact data of the channel, as shown in [6], [11]. This is useful because perfect information is hard or expensive. Moreover, RIS panels are generally sizable and have many elements, resulting in significant training overhead and decreasing the SE. In this direction, in most works regarding the RIS, authors have assumed independent Rayleigh fading, e.g. [10] and [12], but few works have considered RIS correlation, which is inevitable in practice. For example, the performance of RIS correlation has been analyzed in [11] and [12]. Notably, only in [6], [13]–[15], correlated Rayleigh fading has been considered in STAR-RIS-aided systems.

In [11] and [16], the authors proposed a projected gradient ascent method (PGAM) to optimize the RIS phase. However, in [11], authors only employed the PGAM to optimize the phase in an elements-wise form, which needs to be more general and have low efficiency. Authors in [16] expressed the phase in a vector-based format to enhance generality, leading to distinct analytical expressions. However, they did not optimize the amplitudes. In this work, we present a novel PGAM algorithm to optimize both the phases and amplitudes in vector-wise expressions in the STAR-RIS-aided system to improve the transmission throughput further.

In wireless communications, assigning the same frequency band to several users while minimizing interference and boosting data transfer efficiency presents a challenge [17]. Previous works showed that non-orthogonal multiple access (NOMA) is notable for enabling service to multiple users within one resource block using superposition coding. However, NOMA relies on sequential interference elimination, which can cause error accumulation and performance decline with imperfect CSI, as shown in [7], [18]. Traditional broadcasting (BC), while efficient for simultaneous transmission to multiple users, is not suited for individualized data delivery. Rate-splitting (RS) has been proposed as an alternative, dividing the transmit signal into common and private parts. Common messages shared via stream for simultaneous distribution to all users ensure equal access to updates, improving the SE. Private messages, however, use dedicated streams for each user. By allocating both common and private messages, the system achieves a balanced distribution of common information and personal communication. RS offers advantages over both NOMA and BC by reducing error propagation and providing greater flexibility for both individualized and broadcasting-like services. In parallel, the RS strategy has been suggested to tackle multi-user interference [19]–[21]. In this work, we explore the performance benefits of RS over NOMA and BC. Note that the RS has already been used in previous works, e.g., [22] and [23]. It has been confirmed that the RS can improve the SE compared to the traditional BC method [22], [23] and NOMA [20], especially at a high signal-to-noise ratio (SNR). The RS method and STAR-RIS can benefit each other by enhancing the system transmission throughput and

H. Ge, N. Garg and T. Ratnarajah are with Institute for Digital Communications, The University of Edinburgh, Edinburgh, EH9 3FG, UK (e-mails: {hanxiao.ge, navneet.garg, t.ratnarajah}@ed.ac.uk).

A. Papazafeiropoulos is with the Communications and Intelligent Systems Research Group, University of Hertfordshire, Hatfield AL10 9AB, U. K., and with SnT at the University of Luxembourg, Luxembourg (email: tapapazaf@gmail.com).

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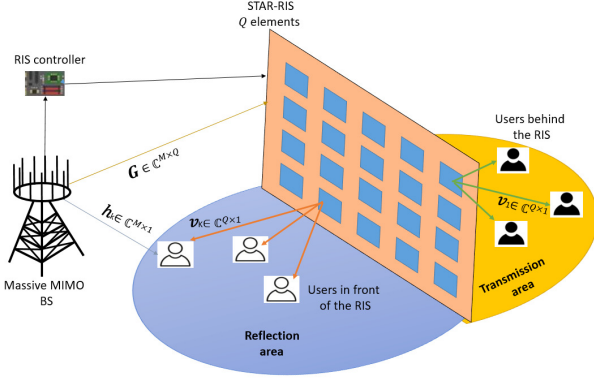


Fig. 1. Structure of a STAR-RIS-aided mMIMO system.

adaptability of mMIMO systems. By combining these two techniques, a more efficient communication system can be achieved, effectively dealing with the challenges posed by complex wireless environments. In previous works, the RS has been employed in RIS-aided systems to enhance the SE, as seen in [16]. However, only a limited number of studies have considered the RS method in STAR-RIS-aided systems, such as [24], [25]. In [24], the authors focused on the outage probability analysis, while herein, we focus on the sum-rate. On the other hand, authors in [25] optimized the phases of the STAR-RIS but still need to address the amplitudes. The optimization of amplitudes can lead to the improvement of the throughput. Additionally, the authors did not consider correlated channels and imperfect CSI.

A. Contributions

Our proposed system is based on a STAR-RIS-aided architecture with imperfect CSI. The RS strategy is applied in this system to improve the signal-to-interference-plus-noise ratio (SINR). In previous works on RIS, such as [22], only the phase shifts were optimized while they did not account for the RIS amplitudes simultaneously, which is less optimal. We use a PGAM algorithm to simultaneously optimize the RIS panel's phase shifts and amplitudes based on statistical CSI, which can get lower overhead compared with other works such as [11] and [26]. We use the ES and the MS protocols as in [27] to adjust the transmission and reflection coefficients. Our contributions are:

- We design a STAR-RIS-aided system with one base station (BS) and multiple users on each side of the surface. We consider that the BS has imperfect CSI, and we carry out the channel estimation by using linear minimum mean-squared error (LMMSE). The asymptotic problem has been obtained for the precoder design when the BS antenna number becomes large.
- We apply the RS strategy in the STAR-RIS-aided downlink system and derive the SINRs for both private and common parts, enabling us to calculate the sum-rate. The benchmark expression without application of the RS strategy is also provided.
- We optimize the sum-rate with the RS strategy through PGAM simultaneously with respect to the amplitudes and phases of the STAR-RIS panel. The sum-rates for the common and private parts are optimized together without being separated. For STAR-RIS, which includes the double number of variables compared to reflecting only RIS, the simultaneous optimization of amplitudes and phase shifts is quite beneficial since it reduces the signal overhead by more effectively utilizing available resources and adapting to channel conditions. We also

give the solution to the optimization problem when we use the MS protocol, which is different from the ES case.

- We compare the system with the RS method with the BC method. Besides, we also consider the differences between the STAR-RIS-aided system with the RIS-aided system. The ES and MS protocols are used to adjust the transmission and reflection coefficients, and we compare the performance of the different protocols. Perfect and imperfect CSI have also been compared in our simulations.

For the rest of this paper, Section II introduces the structure of the STAR-RIS-aided mMIMO system. We also introduce the RS, BC, and NOMA methods. In Section III, we use the LMMSE channel estimation and obtain the imperfect CSI. Section IV presents the SINRs for the private and common parts and the design of the precoding vector. In Section V, the PGAM has been applied in the STAR-RIS-aided system with the RS. Simulations are provided in Section VI. In Section VII, the conclusion is presented.

II. SYSTEM MODEL

We consider a STAR-RIS-aided mMIMO system, where a BS has M antennas that serve K users. One RIS panel with Q elements is equipped in this system. The structure of our proposed system is shown in Fig. 1. A set of users $\mathcal{K}_t = \{1, \dots, K_t\}$ users are located at the transmission region (t) or behind the RIS and $\mathcal{K}_r = \{1, \dots, K_r\}$ users are located in the reflection region (r) or in front of the RIS. We use $\mathcal{W} = \{w_1, w_2, \dots, w_K\}$ to denote the RIS operation mode for each user. If $w_k = t$, the k th user is located behind the RIS panel. In contrast, if $w_k = r$, the k th user is facing the RIS panel.

We assume that $t_q = (\beta_q^t e^{j\phi_q^t}) s_q$ and $r_q = (\beta_q^r e^{j\phi_q^r}) s_q$ are the transmitted and reflected signals by the q th RIS element, where s_q is the impinging signal to the STAR-RIS. In Fig. 1, we use the green line and the orange line to represent t_q and r_q respectively. The amplitude and phase parameters can be given as $\beta_q^{w_k} \in [0, 1]$ and $\phi_q^{w_k} \in [0, 2\pi)$. The choice of the amplitudes is $(\beta_q^t)^2 + (\beta_q^r)^2 = 1, \forall q$.

A. Channel design

In the proposed system, the coherence time equals to $\tau_c = \tau + \tau_d$, where τ is the duration of the uplink training, and τ_d is the duration for the downlink data transmission. The consolidated channel vector for user k , consisting of the direct and cascaded links, which is

$$\mathbf{u}_k = \mathbf{h}_k + \mathbf{G}\Phi_{w_k}\mathbf{v}_k, \quad (1)$$

where $\mathbf{u}_k \in \mathbb{C}^{M \times 1}$ is the consolidated channel of the RIS-aided system, where $\mathbf{R}_k = \mathbb{E}\{\mathbf{u}_k \mathbf{u}_k^H\} \in \mathbb{C}^{M \times M}$; $\Phi_{w_k} = \text{diag}(\beta_1^{w_k} \theta_1^{w_k}, \dots, \beta_Q^{w_k} \theta_Q^{w_k})$ is the phase shifts matrix of the RIS; $\theta_q^{w_k} = e^{j\phi_q^{w_k}}$ is the phase coefficient of the q th element, and $|\theta_q^{w_k}| = 1, \forall q$. Also, $\mathbf{h}_k \in \mathbb{C}^{M \times 1} \sim \mathcal{CN}(\mathbf{0}, \mathbf{R}_{\text{BS}})$ is the direct channel between the BS and user k ; $\mathbf{v}_k \in \mathbb{C}^{Q \times 1} \sim \mathcal{CN}(\mathbf{0}, \mathbf{R}_{\text{RIS}})$ is the channel vector between the RIS and the k th user, where $\mathbf{R}_{\text{RIS}} \in \mathbb{C}^{Q \times Q}$ and $\mathbf{R}_{\text{BS}} \in \mathbb{C}^{M \times M}$ are the correlation matrices at the RIS and the BS; $\mathbf{G} \in \mathbb{C}^{M \times Q}$ is the channel matrix between the BS and the RIS, which is written as [11], [28]

$$\mathbf{G} = \sqrt{\tilde{\beta}_g} \mathbf{R}_{\text{BS}}^{1/2} \mathbf{C} \mathbf{R}_{\text{RIS}}^{1/2}, \quad (2)$$

where $\tilde{\beta}_g$ represents the coefficient of large-scale fading for the channel denoted by $\mathbf{G}$. Also, $\text{vec}(\mathbf{C}) \sim \mathcal{CN}(\mathbf{0}, \mathbf{I}_{MQ})$ is the fast-fading part for $\mathbf{G}$. Meanwhile, $\mathbf{h}_k$ and $\mathbf{v}_k$ are based on correlated Rayleigh fading¹ as

¹The correlated Rician fading will be discussed in future work.

$$\mathbf{h}_k = \sqrt{\beta_k} \mathbf{R}_{\text{BS}}^{1/2} \mathbf{m}_{d,k}, \quad (3)$$

$$\mathbf{v}_k = \sqrt{\eta_k} \mathbf{R}_{\text{RIS}}^{1/2} \mathbf{m}_k, \quad (4)$$

where $\mathbf{m}_{d,k} \sim \mathcal{CN}(0, \mathbf{I}_M)$, $\mathbf{m}_k \sim \mathcal{CN}(0, \mathbf{I}_Q)$ are the corresponding fast-fading vectors. β_k and η_k denote the coefficients of large-scale fading for the channels $\mathbf{h}_k$ and $\mathbf{v}_k$, respectively. In our work, the ensuing analysis relies on the application of statistical CSI. The covariance matrix of $\mathbf{u}_k$ is

$$\begin{aligned} \bar{\mathbf{R}}_k &= \mathbb{E}\{\mathbf{u}_k \mathbf{u}_k^H\} = \mathbb{E}\{\mathbf{h}_k \mathbf{h}_k^H\} + \mathbb{E}\{\mathbf{G} \Phi_{w_k} \mathbf{v}_k \mathbf{v}_k^H \Phi_{w_k}^H \mathbf{G}^H\} \\ &= \tilde{\beta}_k \mathbf{R}_{\text{BS}} + \tilde{\beta}_g \eta_k \mathbf{G} \Phi_{w_k} \mathbf{R}_{\text{RIS}} \Phi_{w_k}^H \mathbf{G}^H \\ &= \tilde{\beta}_k \mathbf{R}_{\text{BS}} + \tilde{\beta}_g \eta_k \text{tr}(\mathbf{R}_{\text{RIS}} \Phi_{w_k} \mathbf{R}_{\text{RIS}} \Phi_{w_k}^H) \mathbf{R}_{\text{BS}} \\ &= \tilde{\beta}_k \mathbf{R}_{\text{BS}} + \tilde{\beta}_g \eta_k \text{tr}(\mathbf{A}_{w_k} \Phi_{w_k}^H) \mathbf{R}_{\text{BS}}, \end{aligned} \quad (5)$$

where $\mathbf{A}_{w_k} = \mathbf{R}_{\text{RIS}} \Phi_{w_k} \mathbf{R}_{\text{RIS}}$, and we have used $\mathbb{E}\{\mathbf{v}_k \mathbf{v}_k^H\} = \eta_k \mathbf{R}_{\text{RIS}}$, and $\mathbb{E}\{\mathbf{h}_k \mathbf{h}_k^H\} = \tilde{\beta}_k \mathbf{R}_{\text{BS}}$.

B. Operation protocols

Our analysis will rely on the ES and MS protocols [27]. Their brief description follows.

1) *ES protocol*: For this protocol, the phase matrix is $\Phi_{w_k}^{\text{ES}} = \text{diag}(\beta_1^{w_k} \theta_1^{w_k}, \dots, \beta_Q^{w_k} \theta_Q^{w_k}) \in \mathbb{C}^{Q \times Q}$, where $\beta_q^{w_k} \geq 0$, $(\beta_q^t)^2 + (\beta_q^r)^2 = 1, \forall q$, and $|\theta_q^{w_k}| = 1, \forall q$.

2) *MS protocol*: For the MS, the related condition is $\Phi_{w_k}^{\text{MS}} = \text{diag}(\beta_1^{w_k} \theta_1^{w_k}, \dots, \beta_Q^{w_k} \theta_Q^{w_k}) \in \mathbb{C}^{Q \times Q}$, where $\beta_q^{w_k} \in \{0, 1\}$, and $(\beta_q^t)^2 + (\beta_q^r)^2 = 1, \forall q$, and $|\theta_q^{w_k}| = 1, \forall q$. The MS is a special condition of the ES protocol, where the amplitude for transmission and reflection are restricted to binary values. In general, the ES performs better than the MS protocol because the MS protocol cannot achieve the full-dimension beamforming gains.

C. Ideal downlink signal

Our system is based on the downlink transmission from the BS to the users through the STAR-RIS panel. The received signal at the k th user is

$$y_k = \mathbf{u}_k^H \mathbf{x} + n_k, \quad (6)$$

where $\mathbf{x} \in \mathbb{C}^{M \times 1}$ is the transmit symbol by the BS, and $n_k \sim \mathcal{CN}(0, \sigma^2)$ is the Gaussian noise with σ being its standard deviation.

D. BC precoding

For the traditional BC precoding [23], we assume that $\mathbf{Z} = [\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_K] \in \mathbb{C}^{M \times K}$ is the precoding matrix of the data symbol s_k .² We can write the transmit signal vector to all users as

$$\mathbf{x}_b = \sqrt{\frac{\lambda P_0}{K}} \sum_{k=1}^K \mathbf{z}_k s_k. \quad (7)$$

We use the normalization parameter λ regarding the precoder to ensure $\mathbb{E}\{\mathbf{x}_b^H \mathbf{x}_b\} = P_0$, where P_0 is the total transmitted power. Hence, λ can be given as

$$\lambda = \frac{K}{\mathbb{E}\{\text{tr}(\mathbf{Z} \mathbf{Z}^H)\}} = \frac{K}{\sum_{i=1}^K \mathbb{E}\{\text{tr}(\mathbf{z}_i \mathbf{z}_i^H)\}}. \quad (8)$$

²The consideration of more sophisticated precoders such as regularized zero-forcing (RZF) will be discussed in future work.

E. NOMA

NOMA is a technique for multiple access that enables several users to simultaneously utilize the same resource block by superposing their signals with different power levels, where the transmitted signal is represented as [29]

$$\mathbf{x}_N = \sum_{k=1}^K \sqrt{\alpha_k \lambda_N P_0} \mathbf{z}_k s_k, \quad (9)$$

where α_k is the proportion of transmitting power to different user k . By allocating different power levels to users, NOMA can achieve efficient spectrum utilization and user multiplexing. The above equation highlights the non-orthogonal resource sharing and power allocation among multiple users, achieved through superposition coding and precoding with $\alpha_1 > \dots > \alpha_K$ [29], [30]. Similarly, to ensure $\mathbb{E}\{\mathbf{x}_N^H \mathbf{x}_N\} = P_0$, the normalization parameter for NOMA is $\lambda_N = \frac{1}{\mathbb{E}\{\text{tr}(\mathbf{Z} \mathbf{Z}^H)\}} = \frac{1}{\sum_{i=1}^K \alpha_i \mathbb{E}\{\text{tr}(\mathbf{z}_i \mathbf{z}_i^H)\}}$, where $\mathbf{E} = \text{diag}(\alpha_1, \dots, \alpha_K)$.

F. RS approach

The RS approach has demonstrated considerable potential in facilitating multi-user transmission under conditions of imperfect CSI, e.g., see [22]. Contrary to the BC strategy, the RS strategy can treat interference in the system as noise and perform interference decoding through the presence of a common message [23]. To minimize interference among multiple users, the RS strategy should be employed to divide the messages into common and private components and to formulate the precoding design. If the power for the common and private parts are ρ_c and ρ , the transmit signal vector is

$$\mathbf{x} = \underbrace{\sqrt{\rho_c} \mathbf{z}_c s_c}_{\text{common part}} + \underbrace{\sqrt{\frac{\lambda \rho}{K}} \sum_{k=1}^K \mathbf{z}_k s_k}_{\text{private part}}, \quad (10)$$

where our focus is on distributing power between common and private messages. We achieve this by implementing uniform power allocation for messages [23]. Hence, we have $\rho_c = P_0(1 - \bar{t})$ with $\bar{t} \in (0, 1]$ being the power-allocation factor and $\rho = P_0 \bar{t}$ being the power allocated to the common and the private part; $\mathbf{z}_c, \mathbf{z}_k \in \mathbb{C}^{M \times 1}$ are the precoder vectors of each parts; s_c and s_k are the messages of each parts. When the RS strategy is applied, the received signal at user k is

$$y_k = \mathbf{u}_k^H \mathbf{x} + n_k \quad (11a)$$

$$= \mathbf{u}_k^H \sqrt{\rho_c} \mathbf{z}_c s_c + \sqrt{\frac{\lambda \rho}{K}} \sum_{k=1}^K \mathbf{u}_k^H \mathbf{z}_k s_k + n_k. \quad (11b)$$

If we insert (10) into (11a), (11b) is obtained.

III. CHANNEL ESTIMATION

In practice, CSI is not perfectly known at the receiver, and channels are estimated during an uplink training phase with pilots. The transmitter obtains some knowledge of the channel given as $\hat{\mathbf{u}}_k$, which is different from the actual channel $\mathbf{u}_k$ according to

$$\hat{\mathbf{u}}_k = \mathbf{u}_k + \tilde{\mathbf{u}}_k, \quad (12)$$

where $\tilde{\mathbf{u}}_k$ is the CSI error vector. Generally, there are two strategies to carry out the channel estimation. In our study, unlike [31] which estimates channels individually, we adopt the method from [11] that estimates end-to-end channels, reducing hardware and power needs. This approach becomes more practical than the first as the number of elements grows. This method is novel in STAR-RIS-aided systems with RS,

allowing a unified channel estimation across user types. We employ the LMMSE method for its efficiency in handling the aggregated channel vector, which is a technique not previously applied in STAR-RIS with RS.

We assume that the uplink training has a duration of τ channel uses. Pilot sequences can be denoted as $\mathbf{x}_{p,k} = [x_{p,k,1}, \dots, x_{p,k,\tau}]^T \in \mathbb{C}^{\tau \times 1}$ with $\mathbf{x}_{p,k}^H \mathbf{x}_{p,j} = 0, \forall j \neq k$ and $\mathbf{x}_{p,k}^H \mathbf{x}_{p,k} = \tau P$, where P is the power for training. During the training, the received signal by the BS is

$$\mathbf{Y}_p = \sum_{i=1}^K \mathbf{u}_i \mathbf{x}_{p,i}^H + \mathbf{N}_p, \quad (13)$$

where $\mathbf{N}_p$ is the Gaussian noise matrix at the BS, which is distributed as $\mathcal{CN}(\mathbf{0}, \sigma^2 \mathbf{I}_M)$. The received training signal by the BS is affected by the noise. By applying the transmitted training sequence $\mathbf{x}_{p,k}$ to counteract interference from other users, we can describe the noisy channel observation for user k as

$$\mathbf{t}_k = \mathbf{u}_k + \frac{1}{\tau P} \mathbf{N}_p \mathbf{x}_{p,k}. \quad (14)$$

The LMMSE channel estimation can be carried out as

$$\hat{\mathbf{u}}_k = \bar{\mathbf{R}}_k \mathbf{Q}_k \mathbf{t}_k, \quad (15)$$

where $\mathbf{Q}_k = (\bar{\mathbf{R}}_k + \frac{\sigma^2}{\tau P} \mathbf{I}_M)^{-1}$.

Proof: Please see Appendix A. ■

Meanwhile, the means of $\hat{\mathbf{u}}_k$ and $\tilde{\mathbf{u}}_k$ are zero, and their variances can be given as

$$\hat{\mathbf{R}}_k = \mathbb{E}\{\hat{\mathbf{u}}_k \hat{\mathbf{u}}_k^H\} = \bar{\mathbf{R}}_k \mathbf{Q}_k \bar{\mathbf{R}}_k \quad (16)$$

$$\tilde{\mathbf{R}}_k = \bar{\mathbf{R}}_k - \hat{\mathbf{R}}_k. \quad (17)$$

The vector $\mathbf{u}_k$ does not follow a Gaussian distribution, $\hat{\mathbf{u}}_k$ and $\tilde{\mathbf{u}}_k$ are not independent, but they are uncorrelated and each of them has a zero mean. In (12), if the error $\tilde{\mathbf{u}}_k$ is large in magnitude, it indicates that the CSI estimation is not accurate. Regarding the variance $\hat{\mathbf{R}}_k$, it provides insights into how the error in channel estimation is distributed across the system. If the errors are widely spread, it suggests that the channel is unpredictable or that the estimation method used is not robust.

IV. PRECODER DESIGN AND ACHIEVABLE RATE

We present the precoder design for the common and private parts in this Section, and the derivation of the achievable SINR. The precoder design results corresponding to the deterministic equivalent SINRs characterize the transmissions of the private and the common messages of user k . Also, the asymptotic rate performance is obtained.

A. Precoder design

We separately examine the precoders for the common and private messages. For private messages, [16] discusses precoding with perfect CSI by using regularized techniques. With imperfect CSI, as [32] suggests, finding optimal precoders requires complex numerical optimization. To simplify, we employ maximum ratio transmission (MRT) by setting $\mathbf{z}_k = \hat{\mathbf{u}}_k$. For common messages, $\mathbf{z}_c$ aims to maximize their rate, represented as $\mathbf{z}_c = \sum_k a_k \hat{\mathbf{u}}_k$, akin to weighted matched beamforming. This leads to a simplified expression $\lambda = K / (\sum_{i=1}^K \text{tr} \hat{\mathbf{R}}_i)$. Although the MRT precoder is not always optimal, we still adopt this scheme due to various reasons such as simplicity, low computational complexity, and ease of implementation.

B. SINR for the private part

From (11b), if we only choose the transmitted signal vector with only the private part, we can get the received signal as

$$\begin{aligned} y_k &= \sqrt{\frac{\lambda \rho}{K}} \mathbf{u}_k^H \mathbf{z}_k s_k + \sqrt{\frac{\lambda \rho}{K}} \sum_{i \neq k} \mathbf{u}_k^H \mathbf{z}_i s_i + n_k \\ &= \underbrace{\sqrt{\frac{\lambda \rho}{K}} \mathbb{E}\{\mathbf{u}_k^H \mathbf{z}_k\} s_k}_{\text{desired signal}} + \underbrace{\sqrt{\frac{\lambda \rho}{K}} \mathbf{u}_k^H \mathbf{z}_k s_k - \sqrt{\frac{\lambda \rho}{K}} \mathbb{E}\{\mathbf{u}_k^H \mathbf{z}_k\} s_k}_{\text{self-interference}} \\ &\quad + \underbrace{\sqrt{\frac{\lambda \rho}{K}} \sum_{i \neq k} \mathbf{u}_k^H \mathbf{z}_i s_i}_{\text{cross-interference}} + \underbrace{n_k}_{\text{noise}}. \end{aligned} \quad (18)$$

The equation of the SINR for the private part is

$$\overline{\text{SINR}}_{p,k} = \frac{|\mathbb{E}\{\mathbf{u}_k^H \mathbf{z}_k\}|^2}{\text{var}(\mathbf{u}_k^H \mathbf{z}_k) + \sum_{i \neq k} \mathbb{E}\{|\mathbf{u}_k^H \mathbf{z}_i|^2\} + \frac{K \sigma^2}{\lambda \rho}} = \frac{S_k}{I_k}, \quad (19)$$

where the numerator is the power of the desired signal denoted as S_k , and the denominator is the summation of the power of all the interference and noise denoted as I_k . We can rewrite the SINR equation in terms of $\bar{\mathbf{R}}_k$ and $\hat{\mathbf{R}}_k$ as

$$\text{SINR}_{p,k} = \frac{\text{tr}^2 \hat{\mathbf{R}}_k}{\sum_{i=1}^K \text{tr} \bar{\mathbf{R}}_i \hat{\mathbf{R}}_i + \frac{K \sigma^2}{\rho} \sum_{i=1}^K \text{tr}(\hat{\mathbf{R}}_i)}, \quad (20)$$

where $\text{SINR}_{p,k}$ is the corresponding deterministic equivalent (DE) of the SINR for the private part.

Proof: Please see Appendix B. ■

C. SINR for the common part

For the common part, the received signal by the k th user during the downlink transmission can be given as

$$y_c = \mathbf{u}_k^H \sqrt{\rho_c} \mathbf{z}_c s_c + \sqrt{\frac{\lambda \rho}{K}} \sum_{i=1}^K \mathbf{u}_k^H \mathbf{z}_i s_i + n_k. \quad (21)$$

The SINR of the common part can be obtained as

$$\overline{\text{SINR}}_{c,k} = \frac{\rho_c |\mathbb{E}\{\mathbf{u}_k^H \mathbf{z}_c\}|^2}{\frac{\lambda \rho}{K} \sum_{i=1}^K \mathbb{E}\{|\mathbf{u}_k^H \mathbf{z}_i|^2\} + \sigma^2} = \frac{S_{c,k}}{I_{c,k}}. \quad (22)$$

$\text{SINR}_{c,k}$ is the corresponding DE of the SINR for the common part and can be derived as

$$\text{SINR}_{c,k} = \frac{\frac{1}{\lambda \rho} a_c^2 \rho_c \text{tr}^2 \hat{\mathbf{R}}_k}{\frac{1}{K} \text{tr}(\bar{\mathbf{R}}_k \hat{\mathbf{R}}_k) + \frac{1}{K} \sum_{i \neq k} \text{tr}(\bar{\mathbf{R}}_i \hat{\mathbf{R}}_i) + \frac{\sigma^2}{\lambda \rho}}, \quad (23)$$

where $S_{c,k}$ and $I_{c,k}$ are the desired signal and interference powers for the common part.

Proof: Please see Appendix C. ■

D. Deterministic equivalent for the achievable rate

The DEs of the SINRs for BC and RS are such that $\overline{\text{SINR}}_k^u - \text{SINR}_k^u \xrightarrow[m \rightarrow \infty]{a.s.} 0, u = \text{BC, RS}$. The DE rate for user k can be obtained using dominated convergence, as in [33], and the principles of continuous mapping theorem, stated as

$$\overline{\mathcal{R}}_k^u - \mathcal{R}_k^u \xrightarrow[m \rightarrow \infty]{a.s.} 0 \quad u = \text{BC, RS, NOMA}, \quad (24)$$

where $\mathcal{R}_k^u$ is the corresponding DE for the achievable rate, which is $\mathcal{R}_k^u = \frac{\tau_c - \tau}{\tau_c} \log_2(1 + \text{SINR}_k^u)$ [34].

E. SINR for common precoder

Our purpose is to obtain the asymptotic of $SINR_c$ when $M \rightarrow \infty$ [35]. For this reason, we rewrite (22) and solve the problem

$$\mathcal{P1} : \max_{\mathbf{z}_c} \min_k \pi_k \cdot |\mathbb{E}\{\mathbf{u}_k^H \mathbf{z}_c\}|^2 \quad s.t. \quad \|\mathbf{z}_c\|^2 = 1, \quad (25)$$

where $\pi_k = \frac{\lambda_c \rho}{\sum_{i=1}^K \mathbb{E}\{|\mathbf{u}_k^H \mathbf{z}_i|^2\} + \sigma^2}$.

To solve this asymptotic problem, we know $\mathbf{z}_c = \sum_k a_k \hat{\mathbf{u}}_k$ and $\mathbb{E}\{\hat{\mathbf{u}}_k \hat{\mathbf{u}}_k^H\} = \hat{\mathbf{R}}_k$. Hence, we can rewrite $|\mathbb{E}\{\mathbf{u}_k^H \mathbf{z}_c\}|^2$ as

$$|\mathbb{E}\{\mathbf{u}_k^H \mathbf{z}_c\}|^2 = |\mathbb{E}\{\hat{\mathbf{u}}_k \sum_k a_k \hat{\mathbf{u}}_k\}|^2 \asymp \frac{1}{M^2} |a_k^2 \text{tr}^2 \hat{\mathbf{R}}_k|. \quad (26)$$

If (26) is inserted into (25), we can rewrite $\mathcal{P1}$ as $\mathcal{P2}$

$$\mathcal{P2} : \max_{a_k} \min_k \pi_k \frac{1}{M^2} |a_k^2 \text{tr}^2 \hat{\mathbf{R}}_k|, \quad s.t. \quad \sum_k a_k^2 = \frac{1}{M}. \quad (27)$$

If $i \neq k$ for all terms, $|\mathbf{z}_c| = 1$, the asymptotically optimal solution of problem $\mathcal{P2}$ is

$$a_k^* = \frac{1}{\sqrt{M \sum_{j=1}^K \frac{\pi_k \text{tr}^2 \hat{\mathbf{R}}_k}{\pi_j \text{tr}^2 \hat{\mathbf{R}}_j}}}, \forall k. \quad (28)$$

V. RIS OPTIMIZATION

In a STAR-RIS-aided system, optimization of the amplitudes and phase shifts is essential for enhancing performance. To achieve this, we utilize infinite resolution phase shifts and perform the optimization by maximizing the sum-rate. This is done through the application of the SINR equations presented in Section IV. In particular, the sum-rate of the private part is

$$\begin{aligned} SE_p &= \sum_k SE_k = \frac{\tau_c - \tau}{\tau_c} \sum_k \log_2(1 + SINR_{p,k}) \\ &= \frac{\tau_c - \tau}{\tau_c} \sum_k \log_2\left(1 + \frac{S_k}{I_k}\right). \end{aligned} \quad (29)$$

The rate of the common part is

$$\begin{aligned} SE_c &= \frac{\tau_c - \tau}{\tau_c} \min_k \{\log_2(1 + SINR_{c,k})\}_{k=1}^K \\ &= \frac{\tau_c - \tau}{\tau_c} \min_k \{\log_2(1 + \frac{S_{c,k}}{I_{c,k}})\}_{k=1}^K, \end{aligned} \quad (30)$$

where we can get (29) and (30) from (7) and (23). We choose the minimum value in (30) to ensure that the common symbol can be decoded by all users. After the problem $\mathcal{P2}$ is solved, the optimization of the system with respect to the amplitudes and phases for the ES can be given as

$$\begin{aligned} \mathcal{P3} : \max_{\boldsymbol{\theta}, \boldsymbol{\beta}} \quad SE &= SE_c + SE_p \\ s.t. \quad (\beta_q^t)^2 + (\beta_q^r)^2 &= 1, \forall q \\ \beta_q^t \geq 0, \beta_q^r \geq 0, |\theta_q^t| &= |\theta_q^r| = 1, \forall q. \end{aligned} \quad (31)$$

Herein, we perform the optimization by using the PGAM. The method is given as Algorithm 1. The convergence is guaranteed due to the transmit power constraint. Also, we use $SE = f(\boldsymbol{\theta}, \boldsymbol{\beta}) = SE_c + SE_p$. The problem $\mathcal{P3}$ is that it is non-convex and involves interdependencies between the optimization variables, specifically the amplitudes and the phase shifts for both transmission and reflection, as noted in [36]. To address this non-convex challenge, we employ the PGAM method, which enables us to secure a locally optimal solution by iteratively moving towards a stationary point, as demonstrated in [37]. At each iteration, the solution

is projected onto the nearest feasible point that adheres to the unit-modulus constraint. This process ensures convergence while maintaining the constraints. In the case of MS, the interval of amplitudes $[0, 1]$ in $\mathcal{P3}$ becomes binary, i.e. $\{0, 1\}$, as $\mathcal{P4}$

$$\begin{aligned} \mathcal{P4} : \max_{\boldsymbol{\theta}, \boldsymbol{\beta}} \quad SE &= SE_c + SE_p \\ s.t. \quad (\beta_q^t)^2 + (\beta_q^r)^2 &= 1, \forall q \\ \beta_q^t \in \{0, 1\}, \beta_q^r \in \{0, 1\}, |\theta_q^t| &= |\theta_q^r| = 1, \forall q. \end{aligned} \quad (32)$$

The binary constraints for β_q^t and β_q^r in $\mathcal{P4}$ are not easy to solve. By rounding the solution of problem $\mathcal{P3}$ to the nearest binary value, we can attain a performance level that is comparatively satisfactory, even though it likes a simplistic approach. A more precise method will be discussed in future work.

A. Optimization of phase and amplitude coefficients

We assume that the constraints $\Theta = \{\boldsymbol{\theta}_t \in \mathbb{C}^{Q \times 1}, \boldsymbol{\theta}_r \in \mathbb{C}^{Q \times 1} \mid |\theta_q^t| = |\theta_q^r| = 1, q = 1, 2, \dots, Q\}$, and $\mathcal{B} = \{\boldsymbol{\beta}_t \in \mathbb{C}^{Q \times 1}, \boldsymbol{\beta}_r \in \mathbb{C}^{Q \times 1}, (\beta_q^t)^2 + (\beta_q^r)^2 = 1, \beta_q^t \geq 1, \beta_q^r \geq 1, q = 1, 2, \dots, Q\}$. The initial constraint is complex due to the need for optimizing both transmission and reflection beamforming. These two forms of passive beamforming are intricately linked through the energy conservation principle. The corresponding iterations can be expressed as

$$\boldsymbol{\theta}^{(q+1)} = P_{\Theta}(\boldsymbol{\theta}^{(q)} + \mu_q \nabla_{\boldsymbol{\theta}} f(\boldsymbol{\theta}^{(q)}, \boldsymbol{\beta}^{(q)})), \quad (33a)$$

$$\boldsymbol{\beta}^{(q+1)} = P_{\mathcal{B}}(\boldsymbol{\beta}^{(q)} + \xi \mu_q \nabla_{\boldsymbol{\beta}} f(\boldsymbol{\theta}^{(q)}, \boldsymbol{\beta}^{(q)})), \quad (33b)$$

where $\nabla_{\boldsymbol{\theta}} f(\boldsymbol{\theta}^{(q)}, \boldsymbol{\beta}^{(q)})$ and $\nabla_{\boldsymbol{\beta}} f(\boldsymbol{\theta}^{(q)}, \boldsymbol{\beta}^{(q)})$ are related to the complex gradients of $f(\boldsymbol{\theta}, \boldsymbol{\beta})$. $P_{\Theta}(\cdot)$ and $P_{\mathcal{B}}(\cdot)$ are the projections onto Θ and $\mathcal{B}$. μ_q and $\xi \mu_q$ are the step sizes for $\boldsymbol{\theta}$ and $\boldsymbol{\beta}$. In this condition, we assume that $\xi > 0$. For equations (33a) and (33b), the step sizes are important to make the projected gradient converge. At each iteration, the step size μ_q is determined using the Armijo-Goldstein backtracking line search method, as in [37]. To find the smallest step size, we define a quadratic approximation of $f(\boldsymbol{\theta}, \boldsymbol{\beta})$ as

$$\begin{aligned} \tilde{Q}_{\mu}(\boldsymbol{\theta}, \boldsymbol{\beta}; \bar{\mathbf{a}}, \bar{\mathbf{b}}) &= f(\boldsymbol{\theta}, \boldsymbol{\beta}) \\ &+ \langle \nabla_{\boldsymbol{\theta}} f(\boldsymbol{\theta}, \boldsymbol{\beta}), \bar{\mathbf{a}} - \boldsymbol{\theta} \rangle - \frac{1}{\mu} \|\bar{\mathbf{a}} - \boldsymbol{\theta}\|_2^2 \\ &+ \langle \nabla_{\boldsymbol{\beta}} f(\boldsymbol{\theta}, \boldsymbol{\beta}), \bar{\mathbf{b}} - \boldsymbol{\beta} \rangle - \frac{1}{\xi \mu} \|\bar{\mathbf{b}} - \boldsymbol{\beta}\|_2^2, \end{aligned} \quad (34)$$

where we define $\langle \bar{\mathbf{a}}, \bar{\mathbf{b}} \rangle = 2\text{Re}(\bar{\mathbf{a}}^H \bar{\mathbf{b}})$ for complex-valued $\bar{\mathbf{a}}$ and $\bar{\mathbf{b}}$ and $\langle \bar{\mathbf{a}}, \bar{\mathbf{b}} \rangle = \bar{\mathbf{a}}^H \bar{\mathbf{b}}$ for non-complex valued $\bar{\mathbf{a}}$ and $\bar{\mathbf{b}}$. The equation of $\tilde{Q}_{\mu}(\boldsymbol{\theta}, \boldsymbol{\beta}; \bar{\mathbf{a}}, \bar{\mathbf{b}})$ is compared with the sum-rate when we use Algorithm 1. We let $L_q > 0$ and $\kappa \in (0, 1)$, then the step size can be written as $\mu_q = L_q \kappa^{m_q}$, where m_q is the smallest nonnegative integer satisfying $f(\boldsymbol{\theta}^{(q+1)}, \boldsymbol{\beta}^{(q+1)}) > \tilde{Q}_{L_q \kappa^{m_q}}(\boldsymbol{\theta}^{(q)}, \boldsymbol{\beta}^{(q)}, \boldsymbol{\theta}^{(q+1)}, \boldsymbol{\beta}^{(q+1)})$, which can be carried out by iterative procedures. The gradient of $f(\boldsymbol{\theta}, \boldsymbol{\beta})$ is related to $\mathbf{r}^*$, where $\mathbf{r} \in \{\boldsymbol{\theta}, \boldsymbol{\beta}\}$. We rewrite the sum-rate equation described by (29) by using the logarithm as

$$SE_p = c_0 \sum_k (\log(S_k + I_k) - \log I_k), \quad (35)$$

where $c_0 = \frac{\tau_c - \tau}{\tau_c \log_2(e)}$.

Theorem 1. The gradients $\nabla_{\boldsymbol{\theta}} f(\boldsymbol{\theta}, \boldsymbol{\beta})$ and $\nabla_{\boldsymbol{\beta}} f(\boldsymbol{\theta}, \boldsymbol{\beta})$ are given in closed-forms by differentiating (29) and (30) as

Algorithm 1 PGAM for the STAR-RIS Optimization

```

1: Initialisation:  $\theta = [\theta_1, \dots, \theta_Q]^T$  and  $\beta = [\beta_1, \dots, \beta_Q]^T$ ;
    $\theta^0, \beta^0, \mu_1 > 0, \kappa \in (0, 1)$ 
2:  $q \leftarrow 1$ 
3: repeat
4:   repeat
5:      $\theta^{(q+1)} = P_\Theta(\theta^{(q)} + \mu_q \nabla_{\theta} f(\theta^{(q)}, \beta^{(q)}))$ 
      $\beta^{(q+1)} = P_\beta(\beta^{(q)} + \xi \mu_q \nabla_{\beta} f(\theta^{(q)}, \beta^{(q)}))$ 
6:   if  $f(\theta^{(q+1)}, \beta^{(q+1)}) \leq \tilde{Q}_{\mu_q}(\theta^{(q)}, \beta^{(q)}, \theta^{(q+1)}, \beta^{(q+1)})$ 
     then
7:      $\mu_q = \mu_q \kappa$ 
8:   end if
9:   until  $f(\theta^{(q+1)}, \beta^{(q+1)}) > \tilde{Q}_{\mu_q}(\theta^{(q)}, \beta^{(q)}, \theta^{(q+1)}, \beta^{(q+1)})$ 
10:   $\mu_{q+1} \leftarrow \mu_q$ 
11:   $q \leftarrow q + 1$ 
12: Until convergence
13: Output:  $\theta^{(q+1)}, \beta^{(q+1)}$ 

```

$$\begin{aligned}
\nabla_{\theta} f(\theta, \beta) &= [\nabla_{\theta^t} f(\theta, \beta)^T, \nabla_{\theta^r} f(\theta, \beta)^T]^T, \\
\nabla_{\theta^{w_k}} f(\theta, \beta) &= \nabla_{\theta^{w_k}} (SE_c + SE_p) \\
&= \frac{\tau_c - \tau}{\tau_c \log 2} \min_k \left\{ \frac{\nabla_{\theta^{w_k}} S_{c,k} I_{c,k} - S_{c,k} \nabla_{\theta^{w_k}} I_{c,k}}{(1 + \text{SINR}_{c,k}) I_{c,k}^2} \right\}_{k=1}^K \quad (36) \\
&+ \frac{\tau_c - \tau}{\tau_c \log 2} \sum_{k=1}^K \frac{\nabla_{\theta^{w_k}} S_k I_k - S_k \nabla_{\theta^{w_k}} I_k}{(1 + \text{SINR}_k) I_k^2},
\end{aligned}$$

where

$$\nabla_{\theta^t} S_k = \begin{cases} v_k \text{diag}(\mathbf{A}_t \text{diag}(\beta^t)), & w_k = t, \\ 0, & w_k = r, \end{cases} \quad (37)$$

$$\nabla_{\theta^r} S_k = \begin{cases} v_k \text{diag}(\mathbf{A}_r \text{diag}(\beta^t)), & w_k = r, \\ 0, & w_k = t, \end{cases} \quad (38)$$

$$\nabla_{\theta^t} I_k = \text{diag}(\tilde{\mathbf{A}}_{kt} \text{diag}(\beta^t)), \quad (39a)$$

$$\nabla_{\theta^r} I_k = \text{diag}(\tilde{\mathbf{A}}_{kr} \text{diag}(\beta^r)), \quad (39b)$$

where we can write

$$v_k = 2\tilde{\beta}_g \eta_k \text{tr} \hat{\mathbf{R}}_k \text{tr} ((\mathbf{Q}_k \bar{\mathbf{R}}_k + \bar{\mathbf{R}}_k \mathbf{Q}_k - \mathbf{Q}_k \bar{\mathbf{R}}_k^2) \mathbf{R}_{\text{BS}}), \quad (40)$$

$$\tilde{\mathbf{A}}_{kl} = \begin{cases} \bar{v}_k \mathbf{A}_l + \sum_{i \in \mathcal{K}_l} \tilde{v}_{ki} \mathbf{A}_l, & w_k = l, \\ \sum_{i \in \mathcal{K}_l} \tilde{v}_{ki} \mathbf{A}_l, & w_k \neq l, \end{cases} \quad (41)$$

where we assume that $l \in \{t, r\}$, and $\bar{v}_k = \tilde{\beta}_g \eta_k \text{tr}(\Psi_k \mathbf{R}_{\text{BS}})$ with $\tilde{v}_{ki} = \tilde{\beta}_g \eta_k \text{tr}(\tilde{\mathbf{R}}_{ki} \mathbf{R}_{\text{BS}})$, where $\Psi_k = \sum_{i=1}^K \hat{\mathbf{R}}_i - 2(\mathbf{Q}_k \bar{\mathbf{R}}_k \hat{\mathbf{R}}_k + \hat{\mathbf{R}}_k \bar{\mathbf{R}}_k \mathbf{Q}_k - \mathbf{Q}_k \bar{\mathbf{R}}_k \hat{\mathbf{R}}_k \bar{\mathbf{R}}_k \mathbf{Q}_k)$, with $\tilde{\mathbf{R}}_{ki} = \mathbf{Q}_i \bar{\mathbf{R}}_i \hat{\mathbf{R}}_k - \mathbf{Q}_i \bar{\mathbf{R}}_i \hat{\mathbf{R}}_k \bar{\mathbf{R}}_i \mathbf{Q}_i + \hat{\mathbf{R}}_k \bar{\mathbf{R}}_i \mathbf{Q}_i$ and $\hat{\mathbf{R}}_k = \bar{\mathbf{R}}_k + \frac{K\sigma^2}{\rho} \mathbf{I}_M$.

To calculate the first term of (36), for the common part we have

$$\nabla_{\theta^t} S_{c,k} = \begin{cases} \sum_i v_{c,ki,1} \text{diag}((\mathbf{A}_t \text{diag}(\beta^t))) \\ + v_{c,k,2} \text{diag}((\mathbf{A}_t \text{diag}(\beta^t))), & w_k, w_i = t, \\ \sum_i v_{c,ki,1} \text{diag}((\mathbf{A}_t \text{diag}(\beta^t))), & w_i = t, w_k = r, \\ v_{c,k,2} \text{diag}((\mathbf{A}_t \text{diag}(\beta^t))), & w_i = r, w_k = t, \\ 0, & w_k, w_i = r, \end{cases} \quad (42)$$

$$\nabla_{\theta^r} S_{c,k} = \begin{cases} \sum_i v_{c,ki,1} \text{diag}((\mathbf{A}_r \text{diag}(\beta^r))) \\ + v_{c,k,2} \text{diag}((\mathbf{A}_r \text{diag}(\beta^r))), & w_k, w_i = r, \\ \sum_i v_{c,ki,1} \text{diag}((\mathbf{A}_r \text{diag}(\beta^r))), & w_i = r, w_k = t, \\ v_{c,k,2} \text{diag}((\mathbf{A}_r \text{diag}(\beta^r))), & w_i = t, w_k = r, \\ 0, & w_k, w_i = t, \end{cases} \quad (43)$$

$$\nabla_{\theta^t} I_{c,k} = \text{diag}(\tilde{\mathbf{A}}_{c,k,t} \text{diag}(\beta^t)), \quad (44a)$$

$$\nabla_{\theta^r} I_{c,k} = \text{diag}(\tilde{\mathbf{A}}_{c,k,r} \text{diag}(\beta^r)), \quad (44b)$$

where we can write $v_{c,ki,1}$, $v_{c,k,2}$ and $\tilde{\mathbf{A}}_{c,k,l}$ as

$$v_{c,ki,1} = \frac{1}{\rho} a_c^2 \rho_c \tilde{\beta}_g \eta_k \text{tr}^2 \hat{\mathbf{R}}_k (\text{tr}((\bar{\mathbf{R}}_i \mathbf{Q}_i^2 \bar{\mathbf{R}}_i + \bar{\mathbf{R}}_i \mathbf{Q}_i) \mathbf{R}_{\text{BS}})), \quad (45)$$

$$\begin{aligned}
v_{c,k,2} &= \frac{2a_c^2 \rho_c}{\rho} \tilde{\beta}_g \eta_k (\sum_i \text{tr} \hat{\mathbf{R}}_i) \text{tr} \hat{\mathbf{R}}_k (\text{tr}(\mathbf{Q}_k \bar{\mathbf{R}}_k - \bar{\mathbf{R}}_k \mathbf{Q}_k^2 \bar{\mathbf{R}}_k \\
&+ \bar{\mathbf{R}}_k \mathbf{Q}_k) \mathbf{R}_{\text{BS}}), \end{aligned} \quad (46)$$

$$\tilde{\mathbf{A}}_{c,k,l} = \begin{cases} \frac{1}{K} \bar{v}_{c,k,1} \mathbf{A}_l + \frac{1}{K} \sum_{i \neq k, i \in \mathcal{K}_l} \tilde{v}_{c,ki,2} \mathbf{A}_l \\ + \sum_{i \in \mathcal{K}_l} \tilde{v}_{c,ki,3} \mathbf{A}_l, & w_k = l, \\ \frac{1}{K} \sum_{i \neq k, i \in \mathcal{K}_l} \tilde{v}_{c,ki,2} \mathbf{A}_l + \sum_{i \in \mathcal{K}_l} \tilde{v}_{c,ki,3} \mathbf{A}_l, & w_k \neq l, \end{cases} \quad (47)$$

where we have $\bar{v}_{c,k,1} = \tilde{\beta}_g \eta_k \text{tr}((\Psi_{c,k,1} + \hat{\mathbf{R}}_k) \mathbf{R}_{\text{BS}})$, $\tilde{v}_{c,ki,2} = \tilde{\beta}_g \eta_k \text{tr}(\Psi_{c,ki,2} \mathbf{R}_{\text{BS}})$, $\tilde{v}_{c,ki,3} = \tilde{\beta}_g \eta_k \text{tr}(\Psi_{c,i,3} \mathbf{R}_{\text{BS}})$, and

$$\Psi_{c,k,1} = \hat{\mathbf{R}}_k + \bar{\mathbf{R}}_k \mathbf{Q}_k \bar{\mathbf{R}}_k - \bar{\mathbf{R}}_k^2 \mathbf{Q}_k^2 \bar{\mathbf{R}}_k + \bar{\mathbf{R}}_k^2 \mathbf{Q}_k, \quad (48a)$$

$$\Psi_{c,ki,2} = \bar{\mathbf{R}}_k \mathbf{Q}_i \bar{\mathbf{R}}_i - \bar{\mathbf{R}}_k \bar{\mathbf{R}}_i \mathbf{Q}_i^2 \bar{\mathbf{R}}_i + \bar{\mathbf{R}}_k \bar{\mathbf{R}}_i \mathbf{Q}_i, \quad (48b)$$

$$\Psi_{c,i,3} = \mathbf{Q}_i \bar{\mathbf{R}}_i - \bar{\mathbf{R}}_i \mathbf{Q}_i^2 \bar{\mathbf{R}}_i + \bar{\mathbf{R}}_i \mathbf{Q}_i. \quad (48c)$$

Similarly, for the amplitudes, the gradient for the sum-rate is

$$\begin{aligned}
\nabla_{\beta} f(\theta, \beta) &= [\nabla_{\beta^t} f(\theta, \beta)^T, \nabla_{\beta^r} f(\theta, \beta)^T]^T, \\
\nabla_{\beta^{w_k}} f(\theta, \beta) &= \nabla_{\beta^{w_k}} (SE_c + SE_p) \\
&= \frac{\tau_c - \tau}{\tau_c \log 2} \min_k \left\{ \frac{\nabla_{\beta^{w_k}} S_{c,k} I_{c,k} - S_{c,k} \nabla_{\beta^{w_k}} I_{c,k}}{(1 + \text{SINR}_{c,k}) I_{c,k}^2} \right\}_{k=1}^K \quad (49) \\
&+ \frac{\tau_c - \tau}{\tau_c \log 2} \sum_{k=1}^K \frac{1}{1 + \text{SINR}_k} \frac{\nabla_{\beta^{w_k}} S_k I_k - S_k \nabla_{\beta^{w_k}} I_k}{I_k^2},
\end{aligned}$$

where we have

$$\nabla_{\beta^t} S_k = \begin{cases} 2v_k \text{Re}\{\text{diag}(\mathbf{A}_t^H \text{diag}(\boldsymbol{\theta}^t))\}, & w_k = t, \\ 0, & w_k = r, \end{cases} \quad (50)$$

$$\nabla_{\beta^r} S_k = \begin{cases} 2v_k \text{Re}\{\text{diag}(\mathbf{A}_r^H \text{diag}(\boldsymbol{\theta}^r))\}, & w_k = r, \\ 0, & w_k = t, \end{cases} \quad (51)$$

$$\nabla_{\beta^t} I_k = 2\text{Re}\{\text{diag}(\tilde{\mathbf{A}}_{kt}^H \text{diag}(\boldsymbol{\theta}^t))\}, \quad (52a)$$

$$\nabla_{\beta^r} I_k = 2\text{Re}\{\text{diag}(\tilde{\mathbf{A}}_{kr}^H \text{diag}(\boldsymbol{\theta}^r))\}. \quad (52b)$$

Similarly, the gradient with respect to the amplitudes can be written as in (53) to (55b)

$$\nabla_{\beta^t} I_{c,k} = 2\text{Re}\{\text{diag}(\tilde{\mathbf{A}}_{c,k,t}^H \text{diag}(\boldsymbol{\theta}^t))\}, \quad (55a)$$

$$\nabla_{\beta^r} I_{c,k} = 2\text{Re}\{\text{diag}(\tilde{\mathbf{A}}_{c,k,r}^H \text{diag}(\boldsymbol{\theta}^r))\}. \quad (55b)$$

Proof: Please see Appendices D and E. ■

The non-convex nature of the optimization problem leads to sub-optimal solutions, i.e., the proposed solution is locally optimal. An exhaustive search yields optimal outcomes, its significant computational complexity and time consumption are notable drawbacks.

B. Complexity Analysis

For the private part, to get $\nabla_{\theta^t} S_k$, we need to calculate the diagonal elements of $\mathbf{A}_t \text{diag}(\beta^t)$. We have $\text{diag}(\mathbf{A}_t \text{diag}(\beta^t)) = \text{diag}(\mathbf{A}_t) \odot \beta^t$, where $\odot$ denotes multiplication performed element by element. Because of $\mathbf{A}_{w_k} = \mathbf{R}_{\text{RIS}} \Phi_{w_k} \mathbf{R}_{\text{RIS}}$, we know that the complexity of $\text{diag}(\mathbf{A}_{w_k})$ is $\mathcal{O}(Q^2)$, and thus, the complexity to compute $\nabla_{\theta^t} S_k$ and $\nabla_{\theta^r} S_k$ is also $\mathcal{O}(Q^2)$. Because $\bar{v}_k$ and $\sum_{i \in \mathcal{K}_l} \bar{v}_{ki}$ need $\mathcal{O}(M^2)$ to compute, the complexities of $\nabla_{\theta^t} I_k$ and $\nabla_{\theta^r} I_k$ are also $\mathcal{O}(M^2)$. Overall, the complexity of the gradients for each iteration for the private part is $\mathcal{O}(K(Q^2 + M^2))$. Similarly, for the common part, the complexities to compute $\nabla_{\theta^t} S_{c,k}$ and $\nabla_{\theta^r} S_{c,k}$ are both $\mathcal{O}(Q^2)$. Also, $\bar{v}_{c,k,1}$, $\sum_{i \neq k, i \in \mathcal{K}_l} \bar{v}_{c,ki,2}$ and $\sum_{i \in \mathcal{K}_l} \bar{v}_{c,ki,3}$ need $\mathcal{O}(M^2)$ to compute. Hence, the complexities of $\nabla_{\theta^t} I_{c,k}$ and $\nabla_{\theta^r} I_{c,k}$ are also $\mathcal{O}(M^2)$. The complexity of the gradients for each iteration for the common part is $\mathcal{O}(Q^2 + M^2)$.

Overall, the whole complexity of $f(\boldsymbol{\theta}, \beta)$, which should be the part with the larger complexity, is $\mathcal{O}(K(Q^2 + M^2))$.

C. Convergence Analysis

The gradient functions $\nabla_{\theta} f(\boldsymbol{\theta}, \beta)$ and $\nabla_{\beta} f(\boldsymbol{\theta}, \beta)$ exhibit Lipschitz continuity, characterized by the Lipschitz constants L_{θ} and L_{β} , respectively. Then, we have

$$\nabla_{\beta^t} S_{c,k} = \begin{cases} 2 \sum_i v_{c,ki,1} \text{Re}\{\text{diag}((\mathbf{A}_t^H \text{diag}(\boldsymbol{\theta}^t)))\} + 2v_{c,k,2} \text{Re}\{\text{diag}((\mathbf{A}_t^H \text{diag}(\beta^t)))\}, & w_k, w_i = t, \\ 2 \sum_i v_{c,ki,1} \text{Re}\{\text{diag}((\mathbf{A}_t^H \text{diag}(\boldsymbol{\theta}^t)))\}, & w_i = t, w_k = r, \\ 2v_{c,k,2} \text{Re}\{\text{diag}((\mathbf{A}_t^H \text{diag}(\beta^t)))\}, & w_i = r, w_k = t, \\ 0, & w_k, w_i = r. \end{cases} \quad (53)$$

$$\begin{aligned} f(\bar{\mathbf{a}}, \bar{\boldsymbol{\beta}}) &\geq f(\boldsymbol{\theta}, \beta) + \langle \nabla_{\theta} f(\boldsymbol{\theta}, \beta), \bar{\mathbf{a}} - \boldsymbol{\theta} \rangle - \frac{1}{L_{\theta}} \|\bar{\mathbf{a}} - \boldsymbol{\theta}\|_2^2 \\ &\quad + \langle \nabla_{\beta} f(\boldsymbol{\theta}, \beta), \bar{\boldsymbol{\beta}} - \beta \rangle - \frac{1}{L_{\beta}} \|\bar{\boldsymbol{\beta}} - \beta\|_2^2 \\ &\geq f(\boldsymbol{\theta}, \beta) + \langle \nabla_{\theta} f(\boldsymbol{\theta}, \beta), \bar{\mathbf{a}} - \boldsymbol{\theta} \rangle - \frac{1}{L_{\max}} \|\bar{\mathbf{a}} - \boldsymbol{\theta}\|_2^2 \\ &\quad + \langle \nabla_{\beta} f(\boldsymbol{\theta}, \beta), \bar{\boldsymbol{\beta}} - \beta \rangle - \frac{1}{L_{\max}} \|\bar{\boldsymbol{\beta}} - \beta\|_2^2, \end{aligned} \quad (56)$$

where we have $L_{\max} = \max(L_{\theta}, L_{\beta})$, the line search procedure steps 5 – 9 in Algorithm 1 terminates in finite iterations because the condition in step 9 should be met $\mu_q < L_{\max}$. The sequence of objectives is increased because of the line search, such as $f(\boldsymbol{\theta}^{(q+1)}, \beta^{(q+1)}) > f(\boldsymbol{\theta}^{(q)}, \beta^{(q)})$. $f(\boldsymbol{\theta}^{(q)}, \beta^{(q)})$ converges because the sets Θ and $\mathcal{B}$ are compact. The algorithm reaches a stationary point of $\mathcal{P}3$, which does not guarantee optimality due to the non-convex nature of $\mathcal{P}3$.

VI. SIMULATIONS

This section presents simulations of the sum-rate of the STAR-RIS-aided system. The coherence time is $\tau_c = 50$ channel uses and the uplink training phase duration is $\tau = 10$ channel uses. The STAR-RIS has $Q = 100$ elements, and the BS is deployed with $M = 64$ antennas. We assume that $K = 4$ users are served by the network with $K_r = 2$ being the number of users in the reflection region and $K_t = 2$ users in the transmission area. The power for uplink training is $P = 20$ dBm. The positions of the BS and the STAR-RIS in xy -coordinates are set at $(x_B, y_B) = (0, 0)$ for the BS and $(x_S, y_S) = (50, 10)$ for the RIS, measured in meters. Also, users in r region are located on a straight line between $(x_S - 0.5d_0, y_S - 0.5d_0)$ and $(x_S + 0.5d_0, y_S - 0.5d_0)$ with equal distances between each two adjacent users, and $d_0 = 20$ m in our simulations. Similarly, users in the t region are located between $(x_S + 0.5d_0, y_S + 0.5d_0)$ and $(x_S - 0.5d_0, y_S + 0.5d_0)$. The carrier frequency is 3 GHz (wavelength $\lambda_0 = 0.1$), and the noise variance is -94 dBm. The size of each RIS element is given by $d_H = d_V = \lambda_0/4$ or $\lambda_0/6$, where d_H and d_V are the dimensions in the horizontal and vertical axes, respectively. The RIS element spacing is $\lambda_0/4$. We model the path loss as $\eta_k = A_S d_{S,k}^{-\alpha}$ with $A_S = d_V^2 = d_H^2$ and $\alpha = 3$, where $d_{S,k}$ is the distance between the STAR-RIS and the user k . Similarly, $\tilde{\beta}_g$ is calculated for the distance $d_{B,S}$ between the RIS and the BS. For direct links, $\tilde{\beta}_k$ is defined as $\tilde{\beta}_k = k_{\lambda} d_{B,k}^{-\alpha}$, where $k_{\lambda} = (\lambda_0/(4\pi))^2$ and $d_{B,k}$ is the distance from the BS to user k . The power-allocation factor is set at $\bar{t} = 0.7$.

A. Perfect and imperfect CSI

In this work, we utilize imperfect CSI due to its alignment with real conditions and the fact that channel estimation errors cannot be disregarded. Nevertheless, we still compare the performance of imperfect CSI with perfect CSI, as demonstrated in Fig. 2. To obtain the perfect CSI scenario, it is necessary to ensure that $\hat{\mathbf{R}}_k = \bar{\mathbf{R}}_k$. With an increase in the total power P_0 , there is a corresponding rise in the sum-rate. It is evident that the system with imperfect CSI results in lower throughput,

$$\nabla_{\beta^r} S_{c,k} = \begin{cases} 2 \sum_i v_{c,ki,1} \text{Re}\{(\text{diag}((\mathbf{A}_r^H \text{diag}(\boldsymbol{\theta}^r)))\} + 2v_{c,k,2} \text{Re}\{(\text{diag}((\mathbf{A}_r^H \text{diag}(\boldsymbol{\beta}^r)))\}, & w_k, w_i = r, \\ 2 \sum_i v_{c,ki,1} \text{Re}\{(\text{diag}((\mathbf{A}_r^H \text{diag}(\boldsymbol{\theta}^r)))\}, & w_i = r, w_k = t, \\ 2v_{c,k,2} \text{Re}\{(\text{diag}((\mathbf{A}_r^H \text{diag}(\boldsymbol{\beta}^r)))\}, & w_i = t, w_k = r, \\ 0, & w_k, w_i = t. \end{cases} \quad (54)$$

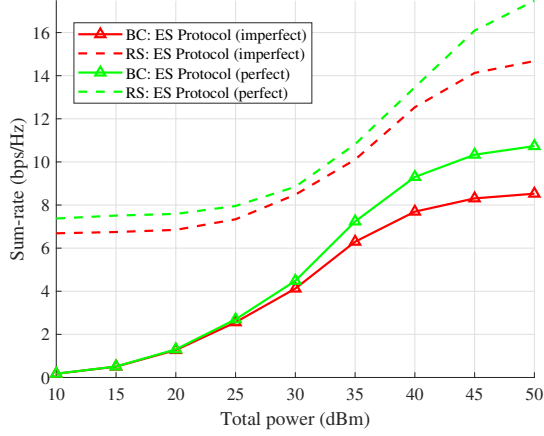


Fig. 2. Sum-rate versus total power P_0 in the STAR-RIS-aided system with perfect and imperfect CSI ($Q = 100, M = 64, K = 4, K_r = 2, d_H = d_V = \lambda_0/4$).

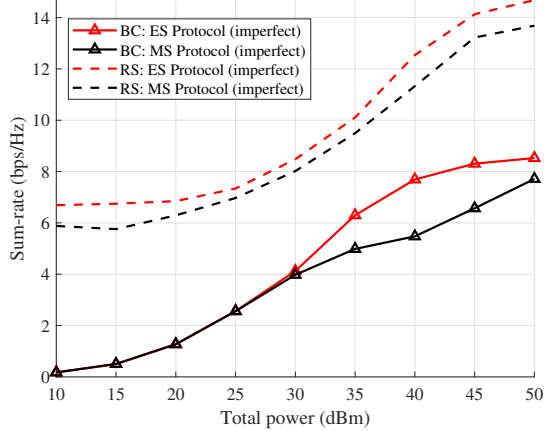


Fig. 3. Sum-rate versus total power P_0 in the STAR-RIS-aided system with different protocols ($Q = 100, M = 64, K = 4, K_r = 2, d_H = d_V = \lambda_0/4$).

particularly for systems employing the traditional BC method. Our results show that the channel estimation error can result in a sum-rate loss of up to 36% in our system.

B. ES and MS protocols

In this work, we evaluate STAR-RIS systems using ES and MS protocols, alongside RS and BC methods. Fig. 3 reveals that combining ES with RS yields the highest sum-rate, since MS is a subset of ES. At low power, both protocols aim for energy efficiency, leading to similar performance. RS notably boosts the sum-rate with ES. However, from 10 dBm to 15 dBm, the sum-rate decreases slightly due to multi-user

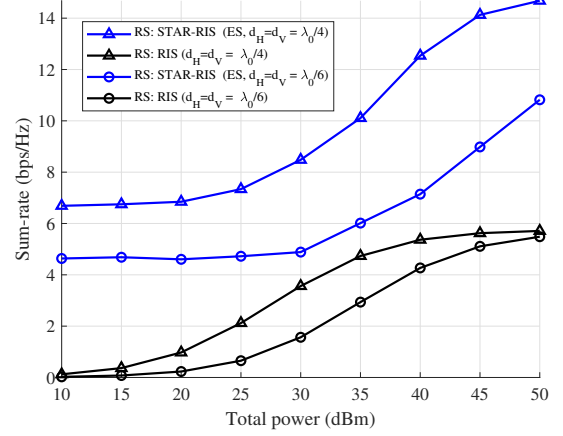


Fig. 4. Sum-rate versus total power P_0 in the STAR-RIS and RIS-aided systems ($Q = 100, M = 64, K = 4, K_r = 2$).

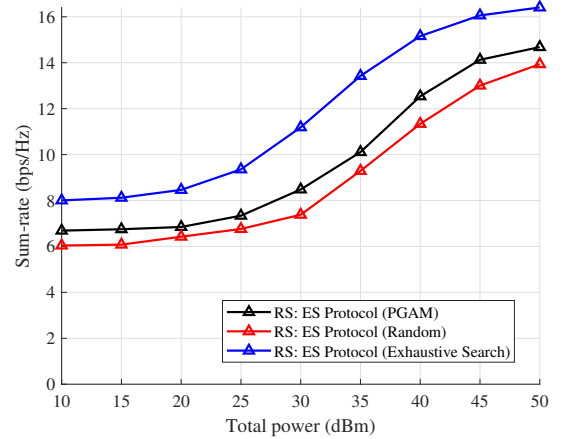


Fig. 5. Sum-rate versus total power P_0 in STAR-RIS-aided systems with different optimization strategies ($Q = 100, M = 64, K = 4, K_r = 2, d_H = d_V = \lambda_0/4$).

interference and less efficient channel resource allocation at lower powers.

C. STAR-RIS-aided and conventional RIS-aided systems

In Fig. 4, we compare the performance of our proposed STAR-RIS-aided system with the ES protocol and the traditional RIS (reflection-only). Then, if we make $\beta_q^r = 1$ and $\theta_q^t = \beta_q^t = 0$, the conventional RIS-aided system can be obtained, which means that $K_t = 2$ users in the transmission region are suspended. However, the rest $K_r = 2$ users in the reflection region are still operating. Our results show that when we choose the STAR-RIS, the sum-rate can be improved compared with the RIS-aided system. In fact, we have also

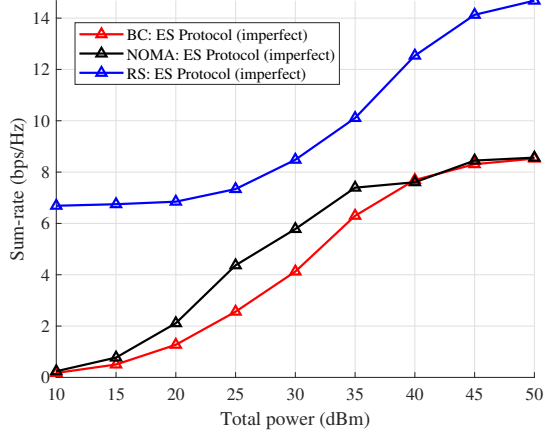


Fig. 6. Sum-rate versus total power P_0 in STAR-RIS-aided systems with the RS/NOMA/BC systems ($K = 4$, $K_r = 2$, $d_H = d_V = \lambda_0/4$).

compared our proposed PGAM algorithm with the case in [37] in the chosen condition. For the RIS-aided system, only phase shifts are optimized as in [37] but for the STAR-RIS case, phases and amplitudes are optimized simultaneously. Our results show that our PGAM algorithm performs better than the algorithm in [37]. Hence, the STAR-RIS should be used to improve the throughput. Also, we compare the system performance when different RIS sizes are chosen. Our results show that when $d_H = d_V = \lambda_0/4$, the sum-rate can be improved because a larger RIS has a better ability to spatially select and reduce the desired signal while suppressing interference from other sources. This can lead to an increase in the transmission throughput.

D. Comparing PGAM with exhaustive search and random selection

In this work, the STAR-RIS-aided system phase shifts and amplitudes are optimized by using PGAM, detailed in Algorithm 1. We compared PGAM with an exhaustive search approach that discretizes each parameter into 10 points, spanning the full range of phase shifts (0 to 2π) and amplitudes (0 to 1), to methodically evaluate all possible configurations. Fig. 5 shows that although exhaustive search yields a sum-rate 11.03% higher than PGAM, the gain does not outweigh its extensive computational demands. PGAM emerges as a viable alternative, slightly underperforming but significantly more efficient, and surpasses random phase/amplitude selection, confirming the efficacy of our method.

E. Comparisons of RS/NOMA/BC methods

In Fig. 6, we compare system throughput using RS, NOMA, and BC transmission methods. RS has the best performance in improving the sum-rate. NOMA excels at low P_0 with specific power allocations ($\alpha_k = [0.31, 0.28, 0.21, 0.2]$) based on user locations and distances from the BS. However, its efficiency declines at high P_0 due to inter-user interference and diminishing returns, nearing BC performance, which uniformly distributes power (P_0/K per user), maintaining throughput even at high P_0 . Thus, at high P_0 , NOMA and BC performances converge.

VII. CONCLUSION

In this paper, we analyzed the performance of a STAR-RIS-aided system with the RS strategy. The transmitter and the receivers are located on both sides of the RIS panels. We considered correlated Rayleigh fading channels, and imperfect CSI

to meet realistic conditions. We applied PGAM to optimize the RIS phase coefficients and amplitudes simultaneously. We compared the system performance with RS, conventional BC, and NOMA. The performances of the STAR-RIS-aided and the traditional RIS-aided systems were also compared. The results confirmed that the ES protocol achieves higher sum-rates than the MS protocol, while the RS outperforms both the BS and NOMA. Additionally, STAR-RIS surpasses traditional RIS in enhancing the sum-rate.

APPENDIX

A. Proof of LMMSE channel estimation

The LMMSE estimator of $\mathbf{u}_k$, which was obtained by minimizing $\text{tr}(\mathbb{E}\{(\hat{\mathbf{u}}_k - \mathbf{u}_k)(\hat{\mathbf{u}}_k - \mathbf{u}_k)^H\})$, is

$$\hat{\mathbf{u}}_k = \mathbb{E}\{\mathbf{t}_k \mathbf{u}_k^H\} (\mathbb{E}\{\mathbf{t}_k \mathbf{t}_k^H\})^{-1} \mathbf{t}_k. \quad (57)$$

Given that the channel and the receiver noise are uncorrelated, we can write

$$\mathbb{E}\{\mathbf{t}_k \mathbf{u}_k^H\} = \mathbb{E}\{\mathbf{u}_k \mathbf{u}_k^H\} = \bar{\mathbf{R}}_k. \quad (58)$$

For the second term of (57), we can write

$$\mathbb{E}\{\mathbf{t}_k \mathbf{t}_k^H\} = \bar{\mathbf{R}}_k + \frac{\sigma^2}{\tau P} \mathbf{I}_M. \quad (59)$$

If we insert (58) and (59) into (15), we can obtain the covariance matrix of the estimated channel vector as

$$\hat{\mathbf{R}}_k = \mathbb{E}\{\hat{\mathbf{u}}_k \hat{\mathbf{u}}_k^H\} = \bar{\mathbf{R}}_k \mathbf{Q}_k \bar{\mathbf{R}}_k. \quad (60)$$

B. Proof of the SINR of the private part

For the numerator, we get $\mathbf{u}_k^H \mathbf{z}_k$ as

$$\mathbb{E}\{\mathbf{u}_k^H \mathbf{z}_k\} = \mathbb{E}\{(\hat{\mathbf{u}}_k + \tilde{\mathbf{u}}_k)^H \hat{\mathbf{u}}_k\} = \mathbb{E}\{\hat{\mathbf{u}}_k^H \hat{\mathbf{u}}_k\} = \text{tr} \hat{\mathbf{R}}_k, \quad (61)$$

where we have used $\mathbf{z}_k = \hat{\mathbf{u}}_k$, and $\mathbb{E}\{\tilde{\mathbf{u}}_k^H \hat{\mathbf{u}}_k\} = 0$, because $\tilde{\mathbf{u}}_k$ and $\hat{\mathbf{u}}_k$ are uncorrelated. Hence, we can get $|\mathbb{E}\{\mathbf{u}_k^H \mathbf{z}_k\}|^2 = \text{tr}^2 \hat{\mathbf{R}}_k$. Regarding the first term of the denominator, we have

$$\text{var}(\mathbf{u}_k^H \mathbf{z}_k) = \mathbb{E}\{|\mathbf{u}_k \hat{\mathbf{u}}_k - \mathbb{E}\{\mathbf{u}_k \hat{\mathbf{u}}_k\}|^2\} \quad (62a)$$

$$= \mathbb{E}\{|\mathbf{u}_k^H \hat{\mathbf{u}}_k| - |\mathbb{E}\{\mathbf{u}_k^H \hat{\mathbf{u}}_k\}|^2\} \quad (62b)$$

$$= \mathbb{E}\{|\hat{\mathbf{u}}_k^H \hat{\mathbf{u}}_k + \tilde{\mathbf{u}}_k^H \hat{\mathbf{u}}_k|^2\} - |\mathbb{E}\{\hat{\mathbf{u}}_k^H \hat{\mathbf{u}}_k\}|^2 \quad (62c)$$

$$= \text{tr}(\bar{\mathbf{R}}_k \hat{\mathbf{R}}_k), \quad (62d)$$

where in (62b), we use $\mathbb{E}\{|C + D|^2\} = \mathbb{E}\{|C|^2\} + \mathbb{E}\{|D|^2\}$. To get (62d), we use $\mathbb{E}\{|\hat{\mathbf{u}}_k^H \hat{\mathbf{u}}_k|^2\} = \text{tr}(\hat{\mathbf{R}}_k^2) + \text{tr}^2(\hat{\mathbf{R}}_k)$. Regarding the second term in the denominator, it is evident that

$$\sum_{i \neq k} \mathbb{E}\{|\mathbf{u}_k^H \mathbf{z}_i|^2\} = \sum_{i \neq k} \mathbb{E}\{\mathbf{u}_k^H \hat{\mathbf{u}}_i \hat{\mathbf{u}}_i^H \mathbf{u}_k\} = \sum_{i \neq k} \text{tr}(\bar{\mathbf{R}}_k \hat{\mathbf{R}}_i). \quad (63)$$

From (8), (62d) and (63), we can derive the SINR for the private part as

$$\text{SINR}_{p,k} = \frac{\text{tr}^2 \hat{\mathbf{R}}_k}{\sum_{i=1}^K \text{tr} \bar{\mathbf{R}}_k \hat{\mathbf{R}}_i + \frac{K\sigma^2}{\rho} \sum_{i=1}^K \text{tr}(\hat{\mathbf{R}}_i)}. \quad (64)$$

C. Proof of the SINR of the common part

For the numerator, we can get

$$\begin{aligned} |\mathbb{E}\{\mathbf{u}_k^H \mathbf{z}_c\}|^2 &= |\mathbb{E}\{\mathbf{u}_k^H \sum_k a_k \hat{\mathbf{u}}_k\}|^2 \\ &= |\mathbb{E}\{(\hat{\mathbf{u}}_k + \tilde{\mathbf{u}}_k) \sum_k a_k \hat{\mathbf{u}}_k\}|^2 = |\mathbb{E}\{\sum_k a_k \hat{\mathbf{u}}_k^H \hat{\mathbf{u}}_k\}|^2 \quad (65) \\ &= a_k^2 \text{tr}^2 \hat{\mathbf{R}}_k, \end{aligned}$$

where $\mathbf{z}_c = \sum_k a_k \hat{\mathbf{u}}_k$ and $\mathbb{E}\{\hat{\mathbf{u}}_k^H \hat{\mathbf{u}}_k\} = \hat{\mathbf{R}}_k$. The summation can be ignored because of the uncorrelation between users. For the denominator, we have

$$\begin{aligned} \frac{1}{K} \sum_{i=1}^K \mathbb{E}\{|\mathbf{u}_k^H \mathbf{z}_i|^2\} &= \frac{1}{K} \sum_{i=1}^K \mathbb{E}\{\mathbf{u}_k^H \mathbf{z}_i \mathbf{z}_i^H \mathbf{u}_k\} \\ &= \frac{1}{K} \sum_{i=1}^K \text{tr}(\bar{\mathbf{R}}_k \hat{\mathbf{R}}_i) = \frac{1}{K} \text{tr}(\bar{\mathbf{R}}_k \hat{\mathbf{R}}_k) + \frac{1}{K} \sum_{i \neq k} \text{tr}(\bar{\mathbf{R}}_k \hat{\mathbf{R}}_i). \quad (66) \end{aligned}$$

Hence, the SINR for the common part is

$$\text{SINR}_{c,k} = \frac{\frac{1}{\lambda_P} a_k^2 \rho_c \text{tr}^2 \hat{\mathbf{R}}_k}{\frac{1}{K} \text{tr}(\bar{\mathbf{R}}_k \hat{\mathbf{R}}_k) + \frac{1}{K} \sum_{i \neq k} \text{tr}(\bar{\mathbf{R}}_k \hat{\mathbf{R}}_i) + \frac{\sigma^2}{\lambda_P}}. \quad (67)$$

D. Proof of the differential of the signal and interference power (private part)

To find $\nabla_{\theta^{w_k}} S_k$ in (36), we should focus on its differentials. Hence, we have

$$d(S_k) = d(\text{tr}^2(\hat{\mathbf{R}}_k)) = 2\text{tr} \hat{\mathbf{R}}_k \text{tr}(d\hat{\mathbf{R}}_k). \quad (68)$$

We know that $\hat{\mathbf{R}}_k = \bar{\mathbf{R}}_k \mathbf{Q}_k \bar{\mathbf{R}}_k$. Hence, we can get the differential of $\hat{\mathbf{R}}_k$ as

$$d(\hat{\mathbf{R}}_k) = d(\bar{\mathbf{R}}_k) \mathbf{Q}_k \bar{\mathbf{R}}_k + \bar{\mathbf{R}}_k d(\mathbf{Q}_k) \bar{\mathbf{R}}_k + \bar{\mathbf{R}}_k \mathbf{Q}_k d(\bar{\mathbf{R}}_k). \quad (69)$$

Because $\mathbf{Q}_k = (\bar{\mathbf{R}}_k + \frac{\sigma^2}{\tau P} \mathbf{I}_M)$, the differential is

$$d(\mathbf{Q}_k) = d((\bar{\mathbf{R}}_k + \frac{\sigma^2}{\tau P})^{-1}) = -\mathbf{Q}_k d(\bar{\mathbf{R}}_k) \mathbf{Q}_k. \quad (70)$$

If we insert (70) into (69), we can get

$$d(\hat{\mathbf{R}}_k) = d(\bar{\mathbf{R}}_k) \mathbf{Q}_k \bar{\mathbf{R}}_k - \bar{\mathbf{R}}_k \mathbf{Q}_k d(\bar{\mathbf{R}}_k) \mathbf{Q}_k \bar{\mathbf{R}}_k + \bar{\mathbf{R}}_k \mathbf{Q}_k d(\bar{\mathbf{R}}_k). \quad (71)$$

From (5), we can derive the derivative of $\bar{\mathbf{R}}_k$ as

$$d(\bar{\mathbf{R}}_k) = \tilde{\beta}_g \eta_k \mathbf{R}_{\text{BS}} \text{tr}(\mathbf{A}_{w_k}^H d(\Phi_{w_k}) + \mathbf{A}_{w_k} d(\Phi_{w_k}^H)). \quad (72)$$

If Φ_{w_k} is diagonal, we can write $d(\bar{\mathbf{R}}_k)$ as

$$\begin{aligned} d(\bar{\mathbf{R}}_k) &= \tilde{\beta}_g \eta_k \mathbf{R}_{\text{BS}} ((\text{diag}(\mathbf{A}_{w_k}^H \text{diag}(\beta^{w_k})))^T d(\theta^{w_k}) \\ &\quad + (\text{diag}(\mathbf{A}_{w_k} \text{diag}(\beta^{w_k})))^T d(\theta^{w_k*})). \end{aligned} \quad (73)$$

Meanwhile, $d(S_k)$ can be written as

$$\begin{aligned} d(S_k) &= 2\text{tr} \hat{\mathbf{R}}_k (\text{tr}(\mathbf{Q}_k \bar{\mathbf{R}}_k (d\bar{\mathbf{R}}_k)) - \text{tr}(\bar{\mathbf{R}}_k \mathbf{Q}_k d(\bar{\mathbf{R}}_k) \mathbf{Q}_k \bar{\mathbf{R}}_k) \\ &\quad + \text{tr}(\bar{\mathbf{R}}_k \mathbf{Q}_k (d\bar{\mathbf{R}}_k))) \end{aligned} \quad (74a)$$

$$\begin{aligned} &= v_k (\text{diag}((\mathbf{A}_{w_k}^H \text{diag}(\beta^{w_k})))^T d\theta^{w_k} \\ &\quad + (\text{diag}(\mathbf{A}_{w_k} \text{diag}(\beta^{w_k})))^T d\theta^{w_k*}), \end{aligned} \quad (74b)$$

where v_k can be obtained as $v_k = 2\tilde{\beta}_g \eta_k \text{tr} \hat{\mathbf{R}}_k \text{tr}((\mathbf{Q}_k \bar{\mathbf{R}}_k + \bar{\mathbf{R}}_k \mathbf{Q}_k - \mathbf{Q}_k \bar{\mathbf{R}}_k^2 \mathbf{Q}_k) \mathbf{R}_{\text{BS}})$. From (74b), we can conclude that $\nabla_{\theta^t} S_k = \frac{\partial S_k}{\partial \theta^{t*}} = v_k \text{diag}(\mathbf{A}_t \text{diag}(\beta^t))$, for $w_k = t$, which can

prove (37). The equation (38) can also be proved if we use the same method. For the private interference, we have

$$\begin{aligned} d(I_k) &= \sum_{i=1}^K \text{tr}(d(\bar{\mathbf{R}}_k) \hat{\mathbf{R}}_i) + \sum_{i=1}^K \text{tr}(\bar{\mathbf{R}}_k d(\hat{\mathbf{R}}_i)) \\ &\quad + \frac{K\sigma^2}{\rho} \sum_{i=1}^K \text{tr}(d(\hat{\mathbf{R}}_i)). \end{aligned} \quad (75)$$

When computing $\nabla_{\theta^t} I_k$, it becomes apparent that $d(\hat{\mathbf{R}}_i) = 0$ due to the condition $w_i \neq t$, which implies that $\hat{\mathbf{R}}_i$ does not depend on θ^t . If $\check{\mathbf{R}}_k = \bar{\mathbf{R}}_k + \frac{K\sigma^2}{\rho} \mathbf{I}_M$, we can rewrite $d(I_k)$ as

$$d(I_k) = \text{tr}(\sum_{i=1}^K \hat{\mathbf{R}}_i d(\bar{\mathbf{R}}_k)) + \sum_{i \in \mathcal{K}_t} \text{tr}(\check{\mathbf{R}}_k d(\hat{\mathbf{R}}_i)). \quad (76)$$

If we use (69) into (76), we can rewrite I_k as

$$\begin{aligned} d(I_k) &= \text{tr}(\sum_{i=1}^K \hat{\mathbf{R}}_i d(\bar{\mathbf{R}}_k)) + \sum_{i \in \mathcal{K}_t} \text{tr}(\check{\mathbf{R}}_k (d(\bar{\mathbf{R}}_i) \mathbf{Q}_i \bar{\mathbf{R}}_k \\ &\quad - \bar{\mathbf{R}}_k \mathbf{Q}_i d(\bar{\mathbf{R}}_i) \mathbf{Q}_i \bar{\mathbf{R}}_i + \bar{\mathbf{R}}_i \mathbf{Q}_i d(\bar{\mathbf{R}}_i))) \\ &= \text{tr}(\Psi_k d(\bar{\mathbf{R}}_k)) + \sum_{i \in \mathcal{K}_t} \text{tr}(\check{\mathbf{R}}_{ki} d(\bar{\mathbf{R}}_i)), \end{aligned} \quad (77)$$

where we can get $\Psi_k = \sum_{i=1}^K \hat{\mathbf{R}}_i$, also $\check{\mathbf{R}}_{ki}$ as $\check{\mathbf{R}}_{ki} = \mathbf{Q}_i \bar{\mathbf{R}}_i \check{\mathbf{R}}_k - \mathbf{Q}_i \bar{\mathbf{R}}_i \check{\mathbf{R}}_k \bar{\mathbf{R}}_i \mathbf{Q}_i + \bar{\mathbf{R}}_k \bar{\mathbf{R}}_i \mathbf{Q}_i$. If $w_k \neq t$, we can get $d(\bar{\mathbf{R}}_k) = 0$, and we can write $\nabla_{\theta^t} I_k$ as $\nabla_{\theta^t} I_k = \frac{\partial I_k}{\partial \theta^{t*}} = \text{diag}(\tilde{\mathbf{A}}_{kt} \text{diag}(\beta^t))$. We assume that $\bar{v}_k = \tilde{\beta}_g \eta_k \text{tr}(\Psi_k \mathbf{R}_{\text{BS}})$ with $\tilde{v}_{ki} = \tilde{\beta}_g \eta_k \text{tr}(\check{\mathbf{R}}_{ki} \mathbf{R}_{\text{BS}})$, where $\tilde{\mathbf{A}}_{kt}$ can be expressed as

$$\tilde{\mathbf{A}}_{kt} = \begin{cases} \bar{v}_k \mathbf{A}_t + \sum_{i \in \mathcal{K}_t} \tilde{v}_{ki} \mathbf{A}_t, & w_k = t, \\ \sum_{i \in \mathcal{K}_t} \tilde{v}_{ki} \mathbf{A}_t, & w_k \neq t. \end{cases} \quad (78)$$

For $w_k = r$, the same method can be used as above. The expression for $\nabla_{\beta^{w_k}} S_k$ can be written as follows. Initially, our focus should be solely on $\nabla_{\beta^t} S_k$, and rewrite $\bar{\mathbf{R}}_k$ as

$$\begin{aligned} d(\bar{\mathbf{R}}_k) &= \tilde{\beta}_g \eta_k \mathbf{R}_{\text{BS}} (\text{diag}(\mathbf{A}_t^H \text{diag}(\theta^t))^T d(\beta^t) \\ &\quad + (\text{diag}(\mathbf{A}_t \text{diag}(\theta^{t*})))^T d(\beta^t)) \\ &= 2\tilde{\beta}_g \eta_k \mathbf{R}_{\text{BS}} \text{Re}\{\text{diag}(\mathbf{A}_t^H \text{diag}(\theta^t))\}^T d(\beta^t). \end{aligned} \quad (79)$$

If we use (79) into (74a), we can get

$$\nabla_{\beta^t} S_k = 2v_k \text{Re}\{\text{diag}(\mathbf{A}_t^H \text{diag}(\theta^t))\} \quad (80)$$

Similarly, $\nabla_{\beta^r} S_k$ can be written as

$$\nabla_{\beta^r} S_k = 2v_k \text{Re}\{\text{diag}(\mathbf{A}_r^H \text{diag}(\theta^r))\} \quad (81)$$

By following the same steps, we can write ∇_{β^t} and ∇_{β^r} as

$$\nabla_{\beta^t} I_k = 2\text{Re}\{\text{diag}(\tilde{\mathbf{A}}_{kt}^H \text{diag}(\theta^t))\}, \quad (82)$$

$$\nabla_{\beta^r} I_k = 2\text{Re}\{\text{diag}(\tilde{\mathbf{A}}_{kr}^H \text{diag}(\theta^r))\}. \quad (83)$$

E. Proof of the differential of the signal and interference powers (common part)

From (67), we can get the derivative of the power $S_{c,k}$ as

$$d(S_{c,k}) = d\left(\frac{1}{\lambda\rho}a_k^2\rho_c tr^2\hat{\mathbf{R}}_k\right) = d\left(\frac{1}{\rho}a_k^2\rho_c\left(\sum_i tr\hat{\mathbf{R}}_i\right)tr^2\hat{\mathbf{R}}_k\right) \quad (84a)$$

$$\begin{aligned} &= \frac{a_k^2\rho_c}{\rho}tr^2\hat{\mathbf{R}}_k\left(\sum_i (tr(\bar{\mathbf{R}}_i\mathbf{Q}_i d(\bar{\mathbf{R}}_i)\mathbf{Q}_i\bar{\mathbf{R}}_i) + tr(\bar{\mathbf{R}}_k\mathbf{Q}_i(d\bar{\mathbf{R}}_i)))\right) \\ &+ \frac{2a_k^2\rho_c}{\rho}\left(\sum_i tr\hat{\mathbf{R}}_i\right)tr\hat{\mathbf{R}}_k(tr(\mathbf{Q}_k\bar{\mathbf{R}}_k(d\bar{\mathbf{R}}_k)) \\ &- tr(\bar{\mathbf{R}}_k\mathbf{Q}_k d(\bar{\mathbf{R}}_k)\mathbf{Q}_k\bar{\mathbf{R}}_k) + tr(\bar{\mathbf{R}}_k\mathbf{Q}_k(d\bar{\mathbf{R}}_k)) \quad (84b) \\ &= \sum_i (v_{c,ki,1}(\text{diag}((\mathbf{A}_{w_i}^H \text{diag}(\beta^{w_i})))^T d\theta^{w_i} \\ &+ (\text{diag}(\mathbf{A}_{w_i} \text{diag}(\beta^{w_i})))^T d\theta^{w_i*})) \\ &+ v_{c,k,2}(\text{diag}((\mathbf{A}_{w_k}^H \text{diag}(\beta^{w_k})))^T d\theta^{w_k} \\ &+ (\text{diag}(\mathbf{A}_{w_k} \text{diag}(\beta^{w_k})))^T d\theta^{w_k*}). \quad (84c) \end{aligned}$$

We can get (84b) from (84a) by using (69). $v_{c,ki,1}$ and $v_{c,k,2}$ are $v_{c,ki,1} = \frac{1}{\rho}a_k^2\rho_c\tilde{\beta}_g\eta_k tr^2\hat{\mathbf{R}}_k(tr((\bar{\mathbf{R}}_i\mathbf{Q}_i^2\bar{\mathbf{R}}_i + \bar{\mathbf{R}}_i\mathbf{Q}_i)\mathbf{R}_{BS}))$ and $v_{c,k,2} = \frac{2a_k^2\rho_c}{\rho}\tilde{\beta}_g\eta_k(\sum_i tr\hat{\mathbf{R}}_i)tr\hat{\mathbf{R}}_k(tr(\mathbf{Q}_k\bar{\mathbf{R}}_k - \bar{\mathbf{R}}_k\mathbf{Q}_k^2\bar{\mathbf{R}}_k + \bar{\mathbf{R}}_k\mathbf{Q}_k)\mathbf{R}_{BS}))$. From (84c), we can get

$$\begin{aligned} \nabla_{\theta^t} S_{c,k} &= \frac{\partial S_{c,k}}{\partial \theta^{t*}} = \sum_i v_{c,ki,1}(\text{diag}((\mathbf{A}_{w_i}^H \text{diag}(\beta^{w_i})))^T d\theta^{w_i} \\ &+ v_{c,k,2}(\text{diag}((\mathbf{A}_{w_k}^H \text{diag}(\beta^{w_k})))^T d\theta^{w_k}, \quad (85) \\ \text{for } w_k = w_i = t, \text{ which can prove (42). Similarly, (43) can also be proved.} \end{aligned}$$

The power of the interference for the common part is

$$\begin{aligned} I_{c,k} &= \frac{1}{K}(tr(\bar{\mathbf{R}}_k\hat{\mathbf{R}}_k) + \sum_{i \neq k} tr(\bar{\mathbf{R}}_k\hat{\mathbf{R}}_i)) + \frac{\sigma^2}{\rho} \sum_{i=1}^K tr\hat{\mathbf{R}}_i \\ &= \frac{1}{K}(I_{c,k,1} + \sum_{i \neq k} I_{c,ki,2}) + \frac{\sigma^2}{\rho} \sum_{i=1}^K I_{c,i,3}. \end{aligned} \quad (86)$$

From (71), we can get

$$\begin{aligned} d(I_{c,k,1}) &= tr(\hat{\mathbf{R}}_k d(\bar{\mathbf{R}}_k)) + tr(\bar{\mathbf{R}}_k d(\hat{\mathbf{R}}_k)) \\ &= tr(\hat{\mathbf{R}}_k d(\bar{\mathbf{R}}_k)) + tr(\bar{\mathbf{R}}_k(d(\bar{\mathbf{R}}_k)\mathbf{Q}_k\bar{\mathbf{R}}_k - \bar{\mathbf{R}}_k\mathbf{Q}_k d(\bar{\mathbf{R}}_k)\mathbf{Q}_k\bar{\mathbf{R}}_k \\ &+ \bar{\mathbf{R}}_k\mathbf{Q}_k(d\bar{\mathbf{R}}_k)) = tr(\Psi_{c,k,1} d(\bar{\mathbf{R}}_k)), \quad (87) \end{aligned}$$

where $\Psi_{c,k,1} = \hat{\mathbf{R}}_k + \bar{\mathbf{R}}_k\mathbf{Q}_k\bar{\mathbf{R}}_k - \bar{\mathbf{R}}_k^2\mathbf{Q}_k^2\bar{\mathbf{R}}_k + \bar{\mathbf{R}}_k^2\mathbf{Q}_k$. For the second part of (86), we have

$$\begin{aligned} d(I_{c,ki,2}) &= tr(\hat{\mathbf{R}}_i d(\bar{\mathbf{R}}_k)) + tr(\bar{\mathbf{R}}_k d(\hat{\mathbf{R}}_i)) \\ &= tr(\hat{\mathbf{R}}_i d(\bar{\mathbf{R}}_k)) + tr(\bar{\mathbf{R}}_k(d(\bar{\mathbf{R}}_i)\mathbf{Q}_i\bar{\mathbf{R}}_i - \bar{\mathbf{R}}_i\mathbf{Q}_i d(\bar{\mathbf{R}}_i)\mathbf{Q}_i\bar{\mathbf{R}}_i \\ &+ \bar{\mathbf{R}}_i\mathbf{Q}_i(d\bar{\mathbf{R}}_i)) = tr(\hat{\mathbf{R}}_i d(\bar{\mathbf{R}}_k)) + tr(\Psi_{c,ki,2} d(\bar{\mathbf{R}}_i)), \quad (88) \end{aligned}$$

where $\Psi_{c,ki,2} = \bar{\mathbf{R}}_k\mathbf{Q}_i\bar{\mathbf{R}}_i - \bar{\mathbf{R}}_k\bar{\mathbf{R}}_i\mathbf{Q}_i^2\bar{\mathbf{R}}_i + \bar{\mathbf{R}}_k\bar{\mathbf{R}}_i\mathbf{Q}_i$. For the third part of (86), we have

$$\begin{aligned} d(I_{c,i,3}) &= tr(d(\bar{\mathbf{R}}_i)\mathbf{Q}_i\bar{\mathbf{R}}_i - \bar{\mathbf{R}}_i\mathbf{Q}_i d(\bar{\mathbf{R}}_i)\mathbf{Q}_i\bar{\mathbf{R}}_i + \bar{\mathbf{R}}_i\mathbf{Q}_i(d\bar{\mathbf{R}}_i)) \\ &= tr(\Psi_{c,i,3} d(\bar{\mathbf{R}}_i)), \quad (89) \end{aligned}$$

where $\Psi_{c,i,3} = \mathbf{Q}_i\bar{\mathbf{R}}_i - \bar{\mathbf{R}}_i\mathbf{Q}_i^2\bar{\mathbf{R}}_i + \bar{\mathbf{R}}_i\mathbf{Q}_i$. Because of $d(\hat{\mathbf{R}}_i) = 0$, the differential of (86) is

$$\begin{aligned} d(I_{c,k}) &= \frac{1}{K}(d(I_{c,k,1}) + \sum_{i \neq k, i \in \mathcal{K}_t} d(I_{c,i,2})) + \frac{\sigma^2}{\rho} \sum_{i \in \mathcal{K}_t} d(I_{c,i,3}) \\ &= \frac{1}{K}(tr(\Psi_{c,k,1} d(\bar{\mathbf{R}}_k)) + tr(\hat{\mathbf{R}}_k d(\bar{\mathbf{R}}_k)) \\ &+ \sum_{i \neq k, i \in \mathcal{K}_t} tr(\Psi_{c,ki,2} d(\bar{\mathbf{R}}_i)) + \frac{\sigma^2}{\rho} \sum_{i \in \mathcal{K}_t} tr(\Psi_{c,i,3} d(\bar{\mathbf{R}}_i)). \quad (90) \end{aligned}$$

If $w_k \neq t$, we get $d(\bar{\mathbf{R}}_k) = 0$, and we can write $\nabla_{\theta^t} I_{c,k}$ as

$$\nabla_{\theta^t} I_{c,k} = \frac{\partial I_{c,k}}{\partial \theta^{t*}} = \text{diag}(\tilde{\mathbf{A}}_{c,k,t} \text{diag}(\beta^t)). \quad (91)$$

We can write $\bar{v}_{c,k,1} = \tilde{\beta}_g\eta_k tr((\Psi_{c,k,1} + \hat{\mathbf{R}}_k)\mathbf{R}_{BS})$, $\tilde{v}_{c,ki,2} = \tilde{\beta}_g\eta_k tr(\Psi_{c,ki,2}\mathbf{R}_{BS})$ and $\tilde{v}_{c,ki,3} = \tilde{\beta}_g\eta_k tr(\Psi_{c,i,3}\mathbf{R}_{BS})$.

$$\tilde{\mathbf{A}}_{c,k,t} = \begin{cases} \frac{1}{K}\bar{v}_{c,k,1}\mathbf{A}_t + \frac{1}{K} \sum_{i \neq k, i \in \mathcal{K}_t} \tilde{v}_{c,ki,2}\mathbf{A}_t \\ + \sum_{i \in \mathcal{K}_t} \tilde{v}_{c,ki,3}\mathbf{A}_t, & w_k = t, \\ \frac{1}{K} \sum_{i \neq k, i \in \mathcal{K}_t} \tilde{v}_{c,ki,2}\mathbf{A}_t + \sum_{i \in \mathcal{K}_t} \tilde{v}_{c,ki,3}\mathbf{A}_t, & w_k \neq t. \end{cases} \quad (92)$$

Like the private part, we consider $\nabla_{\beta^t} S_{c,k}$ if $w_i = w_k = t$ by using (79) into (84b) as

$$\begin{aligned} \nabla_{\beta^t} S_{c,k} &= 2 \sum_i v_{c,ki,1} \text{Re}\{(\text{diag}((\mathbf{A}_t^H \text{diag}(\theta^t)))\} \\ &+ 2v_{c,k,2} \text{Re}\{(\text{diag}((\mathbf{A}_t^H \text{diag}(\beta^t)))\}. \quad (93) \end{aligned}$$

Likewise, we can write $\nabla_{\theta^r} S_{c,k}$ as

$$\begin{aligned} \nabla_{\beta^r} S_{c,k} &= 2 \sum_i v_{c,ki,1} \text{Re}\{(\text{diag}((\mathbf{A}_r^H \text{diag}(\theta^r)))\} \\ &+ 2v_{c,k,2} \text{Re}\{(\text{diag}((\mathbf{A}_r^H \text{diag}(\beta^r)))\}. \quad (94) \end{aligned}$$

For $\nabla_{\beta^t} I_{c,k}$ and $\nabla_{\beta^r} I_{c,k}$, we can follow the same steps as

$$\nabla_{\beta^t} I_{c,k} = 2 \text{Re}\{\text{diag}(\tilde{\mathbf{A}}_{c,k,t}^H \text{diag}(\theta^t))\}, \quad (95)$$

$$\nabla_{\beta^r} I_{c,k} = 2 \text{Re}\{\text{diag}(\tilde{\mathbf{A}}_{c,k,r}^H \text{diag}(\theta^r))\}. \quad (96)$$

Then, we can use (36) and (49) to carry out the joint optimization for both common and private parts simultaneously.

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Hanxiao Ge (Graduate Student Member, IEEE) received a M.Sc. degree in signal processing and communications from The University of Edinburgh, United Kingdom, in 2020. He is currently with Institute for Digital Communications, The University of Edinburgh as a Ph.D. student. His research interests include beyond 5G wireless communications, cell-free massive MIMO systems, and reconfigurable intelligent surfaces.



Anastasios Papazafeiropoulos (Senior Member, IEEE) received the B.Sc. degree (Hons.) in physics, the M.Sc. degree (Hons.) in electronics and computers science, and the Ph.D. degree from the University of Patras, Greece, in 2003, 2005, and 2010, respectively. From 2011 to 2012 and from 2016 to 2017, he was with the Institute for Digital Communications at the University of Edinburgh, U.K., as a Post-Doctoral Research Fellow. From 2012 to 2014, he was a Research Fellow with Imperial College London, U.K., awarded with a Marie Curie fellowship (IEF-IAWICOM). He is currently a Vice-Chancellor Fellow with the University of Hertfordshire, U.K. He

is also a Visiting Research Fellow with the SnT, University of Luxembourg, Luxembourg. He has been involved in several EPSRC and EU FP7 projects such as HIATUS and HARP. His research interests span machine learning for wireless communications, intelligent reflecting surfaces, massive MIMO, heterogeneous networks, 5G and beyond wireless networks, full-duplex radio, mm-wave communications, random matrix theory, hardware-constrained communications, and performance analysis of fading channels.



Navneet Garg (Senior Member, IEEE) received the B.Tech. degree in electronics and communication engineering from College of Science & Engineering, Jhansi, India, in 2010, and the M.Tech. degree in digital communications from ABV-Indian Institute of Information Technology and Management, Gwalior, in 2012. He has completed the Ph.D. degree in June 2018 from the department of electrical engineering at the Indian Institute of Technology Kanpur, India. From July 2018-Jan 2019, he visited The University of Edinburgh, UK. From February 2019-2020, he is employed as a research associate in Heriot-Watt university, Edinburgh, UK. Since February 2020, he

is working as a research associate in The University of Edinburgh, UK. His main research interests include wireless communications, signal processing, optimization, and machine learning.



Tharmalingam Ratnarajah (Senior Member, IEEE) is a Professor of Digital communications and Signal Processing at the Institute for Digital Communications, University of Edinburgh. His research interests include signal processing and information-theoretic aspects of beyond 5G cellular networks, full-duplex radio, mmWave communications, random matrix theory, big data analytics and machine learning for wireless networks, statistical and array signal processing, physical-layer secrecy and interference alignment. He has published over 450 Peer-reviewed papers in these areas and holds four US patents. He was the lead coordinator of the European

Union (EU) projects HARP (4.6M€) in the area of highly distributed MIMO and ADEL (3.7M€) in the area of licensed shared access. He was also the coordinator of the European Union Future and Emerging Technologies project CROWN (3.4M€) in the area of cognitive radio networks and HIATUS (3.6M€) in the area of interference alignment. Prof Ratnarajah was an associate editor of IEEE Transactions on Signal Processing, 2015-2017, and Technical co-chair of the 17th IEEE International Workshop on Signal Processing Advances in Wireless Communications, Edinburgh, July 2016. Prof. Ratnarajah is a Fellow of the Higher Education Academy (FHEA).